LOTTERY PREPARATION PRESERVING HIGHER-ORDER PI-TWO-ONE INDESCRIBABILITY THROUGH FULL HECHLER ITERATIONS

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Editorial note. Complete original mathematical body and embedded bibliography retained. Author takes responsibility for mathematical correctness; no proof repair, additional assumptions or independent certification.

Editorial note. Original declaration2714–2722 retained: Codex Sol6.1 assisted literature, strategy exploration, mathematical argument development and proof checks, organization and preparation; author takes full responsibility. No explicit core-proof AI-discovery target is asserted, and no AI-free/pure-human claim is added.

Editorial note. Full preservation appendix explicitly included in the relevant proof bundles; the higher-order Pi^2_1 convention, local lottery preparation and continuum assumptions remain unchanged.

Editorial note. Three native statements are collected once each; direct capture/iteration corollaries and the alternative stronger-cardinal consistency route are not separately counted.

Allowed standard prerequisites (editorial).

- Higher-order indescribability, weak compactness, generalized Hechler forcing and elementary embedding lifts. Native Pi^2_1 convention is explicitly distinguished from Pi^1_2; all C-sequence amenability definitions, support/localization bounds, canonical presentation, target-model agreement and local lottery preparation assumptions retained.. Complete original body and bibliography, including the full nonadjacent preservation appendix2432–2673; no preservation assumption replaced by a stronger one.

1. Introduction

Conjecture 4.13 of [8] asks whether a regular uncountable cardinal which carries no amenable C-sequence must be ineffable in L. We give a relative consistency counterexample in which the cardinal remains Π^2_1-indescribable.

Definition 1.1 (Notation and C-sequences)**.** Throughout the article, inaccessibility means strong inaccessibility, and $\mathrm{Inacc}^M(\kappa)$ denotes the set of inaccessible cardinals below κ as computed in the specified universe M. Without a superscript it is computed in the current universe. For $X \subseteq \eta$, where η is a limit ordinal, write $\mathrm{acc}(X) = \{\delta < \eta : 0 < \delta = \sup(X \cap \delta)\}$, with the ambient ordinal η understood from context. For a regular uncountable cardinal κ, a *C-sequence on $S \subseteq \kappa$* is a sequence $\vec{C} = \langle C_\beta : \beta \in S\rangle$ such that C_β is club in β at every nonzero limit index. At other indices in its domain we require $C_0 = \varnothing$ and $C_{\alpha+1} = \{\alpha\}$.

Definition 1.2 (Amenability and capture)**.** Let κ be regular uncountable and let $\vec{C}$ be a C-sequence on $S \subseteq \kappa$. For a club $D \subseteq \kappa$, set

$$\mathrm{Cap}(D, \vec{C}) = \{\beta \in S : D \cap \beta \subseteq C_\beta\}.$$

The sequence is *amenable in* κ if this set is nonstationary in κ for every club $D \subseteq \kappa$. We say that D *captures* $\vec{C}$ if $\mathrm{Cap}(D, \vec{C})$ is stationary in κ. Following [8, Definitions 4.1 and 4.3], write

$$\mathcal{A}_\kappa = \{S \subseteq \kappa : S \text{ carries a } C\text{-sequence amenable in } \kappa\}.$$

The notion of amenability was made explicit in [4]; we use the equivalent formulation recorded in [8, Definition 4.1 and Fact 4.2]. The entries at zero and successor ordinals do not affect amenability, since the nonzero limit ordinals form a club. The family $\mathcal{A}_\kappa$ is a κ-complete ideal containing the nonstationary ideal. Restrictions preserve amenability. For a union of fewer than κ sets carrying amenable sequences, assign each index to the first set containing it in a fixed enumeration, and use that set's sequence there.

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For any club, the capture set of the resulting sequence is contained in the union of the original capture sets, a union of fewer than κ nonstationary sets. A sequence on a nonstationary domain is automatically amenable. In particular, $\kappa \notin \mathcal{A}_\kappa$ says exactly that every C-sequence on κ is captured by some club.

Theorem 1.3. *If* ZFC+ *"there is a Π^2_1-indescribable cardinal" is consistent, then so is*

$$\mathsf{ZFC} + \exists \kappa \left[\kappa \text{ is } \Pi^2_1\text{-indescribable } \wedge\ \kappa \notin \mathcal{A}_\kappa\ \wedge\ L \models \kappa \text{ is not ineffable}\right].$$

The converse consistency implication follows from the indescribability asserted in the conclusion, so the two theories in Theorem 1.3 are equiconsistent. Theorem 1.3 also gives a negative answer to [1, Question 5.5], which asks whether the existence of a cardinal carrying no amenable C-sequence is equiconsistent with the existence of an ineffable cardinal. Indeed, by [7, Lemma 5.4], every ineffable cardinal δ is a limit of totally indescribable cardinals. In particular, V_δ satisfies ZFC+ "there is a Π^2_1-indescribable cardinal". Thus the consistency upper bound supplied by our construction is strictly below an ineffable cardinal. We do not determine the optimal consistency strength when the indescribability requirement in the conclusion is omitted. The superscript in Π^2_1 is significant: we use higher-order indescribability, as specified in Definition 2.1, rather than Π^1_2-indescribability.

Models with no amenable C-sequence were obtained by Radin forcing [1], and later by overlapping-extender forcing [2]. We obtain such a model by Hechler forcing, while preserving Π^2_1-indescribability and non-ineffability in L. We prove stationary local domination at weakly compact cardinals.

A related result in [11] gives a model in which all C-sequences are trivial in the sense of the C-sequence number, while a κ-Souslin tree exists. Triviality allows an initial segment of a cofinal set to be covered by an entry chosen at an index that need not equal the endpoint of that initial segment; capture requires a club D such that $D \cap \beta \subseteq C_\beta$ for stationarily many β. Our model retains weak compactness, and every C-sequence is captured in the sense of Definition 1.2.

We start in L, fix its least Π^2_1-indescribable cardinal κ, and show that κ is not ineffable there. We then apply the local lottery preparation of [9] and iterate the full generalized Hechler forcing with support $< \kappa$ for κ^+ stages. Each final C-sequence appears in an intermediate extension; the next Hechler stage captures it, and the closed tail preserves the stationary capture set.

At height $\kappa + 1$, [9, Corollary 38] covers orders of size at most κ. Under $2^\kappa = \kappa^+$, Theorem 5.1 extends this result to the full Hechler iteration. Its proof in Appendix A uses a canonical presentation with conditions in H_{κ^+}, shows that the prepared universe and the embedding target compute the same canonical iteration, and carries out two lifts. For a Π^3_1-indescribable cardinal, the corresponding local preparation has the larger size bound $\beth_{\kappa+1} = 2^\kappa$, so Johnstone's corollary applies directly to a dense presentation of the iteration. Proposition 5.3 and Corollary 5.5 give this alternative argument.

The use of a prepared cardinal followed by a long Hechler iteration also appears in [12, Theorem 4.7], for a conclusion about cardinal characteristics. There, indestructible strong unfoldability is followed by a $< \kappa$-support Hechler iteration of length κ^{++}. Here we use the weaker local hypothesis Π^2_1-indescribability and length κ^+. The capture theorem ensures that every C-sequence in the final extension is non-amenable.

Another use of Hechler domination at stationary sets of indices appears in the proof of [10, Lemma 20]. In that construction, a classical Hechler real dominates functions in ${}^\omega\omega$ associated with stationarily many indices below ω_1; a subsequent forcing adds a club through those indices. Our local domination theorem applies to every ground-model family $\langle f_\beta : \beta \in \mathrm{Inacc}(\kappa)\rangle$, without coherence assumptions on its entries.

The connection between Hechler forcing and club capture is given by next-point functions. For inaccessible β and $\xi < \beta$, put $n_\beta(\xi) = \min(C_\beta \setminus (\xi + 1))$. If $d[\beta] \subseteq \beta$ and $n_\beta \leq^* d \restriction \beta$ hold at stationarily many inaccessible indices, the pressing-down lemma makes the domination threshold uniform on a stationary subset. A single tail of the club of limit closure points of d then captures $\vec{C}$, as shown in Corollary 3.12. Ordinary eventual domination of ground-model functions from κ to κ does not by itself establish this local property: extending n_β to domain κ may place the eventual domination threshold beyond β. The reflection argument at a weakly compact cardinal supplies the stationary local domination needed in Theorem 3.10.

2. Indescribability and the initial cardinal

Definition 2.1 (Indescribability conventions)**.** We use the classical notation Π^m_n, in which the upper index refers to quantifiers of order $m+1$, with full semantics in the ambient universe. Thus Π^1_1 sentences have a universal second-order block and a first-order matrix; Π^2_1 sentences permit a leading universal

block of third-order quantifiers. In particular, Π^2_1 and Π^1_2 are different notions. For inaccessible κ, Π^{m+1}_1-indescribability is equivalent to $(\kappa+m)$-strong unfoldability for $1 \leq m < \omega$, by [9, Corollary 35]. Here $\kappa + m$ uses ordinal addition. The two preservation arguments below use heights $\kappa+1$ and $\kappa+2$, respectively.

We use weak compactness in its reflection form: if $R \subseteq V_\kappa$ and a Π^1_1 sentence holds in $(V_\kappa, \in, R)$, then it holds in $(V_\beta, \in, R \cap V_\beta)$ for some $\beta < \kappa$. Every Π^2_1-indescribable cardinal, and hence every Π^3_1-indescribable cardinal, is weakly compact.

Proposition 2.2. *The least Π^2_1-indescribable cardinal, if it exists, is neither subtle nor ineffable.*

Proof. If the least such cardinal κ were subtle, [7, Lemma 5.4] would give a totally indescribable cardinal $\gamma < \kappa$. In particular, γ would be Π^2_1-indescribable, a contradiction. Since ineffability implies subtlety, κ is not ineffable. □

Corollary 2.3. *The consistency of a Π^2_1-indescribable cardinal implies the consistency of a universe $W \models \mathsf{ZFC} + V = L$ containing such a cardinal.*

Proof. The indescribability hypotheses used in [9] are, if consistent with ZFC, consistent with $V = L$; see [9, Section 1, p. 1215]. □

In that universe we take the least such κ. Proposition 2.2 makes it non-ineffable in L, and GCH gives $2^\kappa = \kappa^+$.

3. Stationary capture by generalized Hechler forcing

We prove local domination for a ground-model family of functions indexed by the inaccessible cardinals below a weakly compact cardinal. We first reflect predensity from $\mathbb{D}_\kappa$ to $\mathbb{D}_\beta$ for an inaccessible $\beta < \kappa$, then construct a condition in $\mathbb{D}_\kappa$ with stem length β. Applied to the next-point functions of a C-sequence, this gives stationary capture.

Definition 3.1 (Generalized Hechler forcing)**.** For a regular uncountable cardinal κ, let $\mathbb{D}_\kappa$ be generalized Hechler forcing in the presentation of [3, Definition 48]. Write its conditions as $p = (s_p, h_p)$, where $s_p \in {}^{<\kappa}\kappa$, $h_p \in {}^{\kappa}\kappa$, and $s_p \subseteq h_p$. We call s_p the *stem*, put $\ell(p) = \mathrm{dom}(s_p)$, and call h_p the *side function*. Thus $q \leq p$ means $s_p \subseteq s_q$ and $h_p \leq h_q$ pointwise. The definition makes sense for every regular uncountable κ; the hypothesis $2^{<\kappa} = \kappa$ in the reference is needed for the chain condition. If $\beta \leq \kappa$ is regular uncountable and $p \in \mathbb{D}_\kappa$ satisfies $\ell(p) < \beta$ and $h_p[\beta] \subseteq \beta$, put

$$p \restriction \beta = (s_p, h_p \restriction \beta).$$

Then $p \restriction \beta \in \mathbb{D}_\beta$. We use this notation only under the stated hypotheses.

Definition 3.2 (Domination and the generic function)**.** For ordinal-valued functions f, g on a nonzero limit ordinal η, write $f \leq^* g$ if

$$(\exists \gamma < \eta)(\forall \xi \in [\gamma, \eta))\, f(\xi) \leq g(\xi).$$

If $G \subseteq \mathbb{D}_\kappa$ is generic, its *Hechler generic function* is $d_G = \bigcup_{p \in G} s_p$. In forcing statements, d denotes the canonical name for this function.

Definition 3.3 (Predensity)**.** For a forcing order $\mathbb{P}$, let $\mathrm{Comp}_{\mathbb{P}}(p, q)$ mean that $p, q \in \mathbb{P}$ are compatible. A family $A \subseteq \mathbb{P}$ is *predense below p* if every $q \leq p$ is compatible with a member of A; members of A need not themselves extend p. We omit the subscript on Comp when the order is clear.

Remark 3.4. For $p = (s, h), q = (t, k) \in \mathbb{D}_\kappa$,

$$\begin{aligned} \mathrm{Comp}_{\mathbb{D}_\kappa}(p, q) \quad \Longleftrightarrow \quad & \big[s \subseteq t \ \wedge\ (\forall \xi < \mathrm{dom}(t))\, h(\xi) \leq t(\xi)\big] \\ & \vee \big[t \subseteq s \ \wedge\ (\forall \xi < \mathrm{dom}(s))\, k(\xi) \leq s(\xi)\big]. \end{aligned} \tag{1}$$

A common extension must have a stem extending both s and t, so the stems are comparable and its values on the longer stem give the corresponding inequality. Conversely, the longer stem and the pointwise maximum of the side functions form a common extension; the inequality ensures normalization. All these tests use only coordinates below the length of the longer stem. Consequently, if both restrictions to a regular uncountable $\beta \leq \kappa$ are defined, then

$$\mathrm{Comp}_{\mathbb{D}_\kappa}(p, q) \quad \Longleftrightarrow \quad \mathrm{Comp}_{\mathbb{D}_\beta}(p \restriction \beta, q \restriction \beta).$$

Lemma 3.5. *For a regular uncountable κ, if $G \subseteq \mathbb{D}_\kappa$ is generic, then d_G is a function from κ to κ that eventually dominates every function f from κ to κ in the ground universe. Moreover,*

$$G = \{p \in \mathbb{D}_\kappa : s_p \subseteq d_G \text{ and } h_p \leq d_G\},$$

where $\mathbb{D}_\kappa$ on the right is the ground-model order. In particular, $V[G] = V[d_G]$.

Proof. Totality and eventual domination follow from the usual density argument; see [3, Section 4.2]. These facts require only regularity of κ. We verify the characterization of G. For $p \in G$ and $\xi < \kappa$, choose a common extension in G of p and a condition whose stem includes ξ. Its value at ξ is $d_G(\xi)$ and is at least $h_p(\xi)$. This proves $s_p \subseteq d_G$ and $h_p \leq d_G$. Conversely, suppose a ground-model condition p satisfies these requirements. For every $q \in G$, the stems of p and q are comparable, since both are initial segments of d_G. Since both side functions are bounded by d_G, the inequality in (1) corresponding to the longer stem holds. Hence p is compatible with every member of G. The dense set of conditions extending p or incompatible with p is met by G, necessarily by a condition extending p. Therefore $p \in G$, proving the displayed characterization. □

Lemma 3.6. *If κ is strongly inaccessible, then $\mathbb{D}_\kappa$ is κ^+-cc* [3, Section 4.2].

Lemma 3.7. *For every regular uncountable κ, the order $\mathbb{D}_\kappa$ is $< \kappa$-closed.*

Proof. For later use, recall the lower bound witnessing closure in [3, Section 4.2]: for a decreasing sequence $\langle (s_i, h_i) : i < \mu \rangle$, with $0 < \mu < \kappa$, put

$$s = \bigcup_{i<\mu} s_i, \qquad h(\xi) = \sup_{i<\mu} h_i(\xi) \quad (\xi < \kappa).$$

Regularity bounds the domain and values below κ, and normalization fixes $h_i(\xi)$ once ξ enters a stem. Thus (s, h) is a normalized lower bound. The empty sequence has the greatest condition $(\varnothing, \mathbf{0})$ as a lower bound. □

For local domination we need predensity in $\mathbb{D}_\beta$, including conditions whose side functions are arbitrary maps from β to β. The next lemma obtains this by Π^1_1 reflection. In the proof of Theorem 3.10, the auxiliary ordinals $\delta_{\alpha,i}$ will be chosen so that $a_{\alpha,i} \Vdash_{\mathbb{D}_\kappa} \delta_{\alpha,i} \in \dot{E}$.

Lemma 3.8. *Let κ be weakly compact, let $p \in \mathbb{D}_\kappa$, and let $\langle A_\alpha : \alpha < \kappa \rangle$ be a sequence of maximal antichains below p. Choose surjections $e_\alpha : \kappa \to A_\alpha$, and write*

$$a_{\alpha,i} = e_\alpha(i) = (s_{\alpha,i}, h_{\alpha,i}) \qquad (\alpha, i < \kappa).$$

Let $\delta : \kappa \times \kappa \to \kappa$ be any function, and write $\delta_{\alpha,i} = \delta(\alpha, i)$. The set of strongly inaccessible $\beta < \kappa$ satisfying the following conditions is stationary in κ:

(i) *$\ell(p) < \beta$ and $h_p[\beta] \subseteq \beta$;*
(ii) *for all $\alpha, i < \beta$, $\ell(a_{\alpha,i}) < \beta$, $\delta_{\alpha,i} < \beta$, and $h_{\alpha,i}[\beta] \subseteq \beta$;*
(iii) *for every $\alpha < \beta$, the family*

$$A^\beta_\alpha = \{a_{\alpha,i} \restriction \beta : i < \beta\}$$

is predense below $p \restriction \beta$ in $\mathbb{D}_\beta$.

Proof. Since κ is strongly inaccessible, Lemma 3.6 gives $1 \leq |A_\alpha| \leq \kappa$. Thus the surjections exist, with repetitions allowed. To prove stationarity, fix a club $B \subseteq \kappa$. Let K be the set of uncountable fixed points of the beth function below κ, and put $K' = K \cap B$. Strong inaccessibility gives $\beth_\xi < \kappa$ for $\xi < \kappa$. The beth function is increasing and continuous, so its uncountable fixed points form a club in κ. Every member of K is an uncountable strong-limit cardinal; we require regularity through a separate reflection clause.

Use finite tagged tuples to form the following predicate $R \subseteq V_\kappa$:

$$R_0 = \{\langle 0, \ell(p)\rangle\}, \qquad R_4 = \{\langle 4, \alpha, i, \xi, h_{\alpha,i}(\xi)\rangle : \alpha, i, \xi < \kappa\},$$

$$R_1 = \{\langle 1, \xi, h_p(\xi)\rangle : \xi < \kappa\}, \qquad R_5 = \{\langle 5, \xi\rangle : \xi \in K'\},$$

$$R_2 = \{\langle 2, \alpha, i, \ell(a_{\alpha,i})\rangle : \alpha, i < \kappa\}, \quad R_6 = \{\langle 6, s_p\rangle\},$$

$$R_3 = \{\langle 3, \alpha, i, \delta_{\alpha,i}\rangle : \alpha, i < \kappa\}, \qquad R_7 = \{\langle 7, \alpha, i, s_{\alpha,i}\rangle : \alpha, i < \kappa\},$$

$$R = \bigcup_{j<8} R_j.$$

Each stem belongs to V_κ, since its domain and range are bounded below the regular cardinal κ. Side functions need not belong to V_κ, so we represent their values by sections of the predicate.

We will apply reflection to the sentence

$$\Phi(R) := \bigwedge_{i=1}^{11} \phi_i.$$

We describe the formulas $\phi_1, \dots, \phi_{11}$ below and then verify that $\Phi(R)$ is Π^1_1. In these formulas, ordinal variables range over the ordinals of the rank structure. The symbols $\ell_p, h_p, \ell_{\alpha,i}, \delta_{\alpha,i}, h_{\alpha,i}, s_p, s_{\alpha,i}$ are read from sections $R_0, R_1, R_2, R_3, R_4, R_6, R_7$, respectively; each is an abbreviation for a first-order formula using R. We use the usual first-order abbreviations for ordinals, functions, and ordered pairs. In particular, $s : \eta \to \mathrm{Ord}$ asserts that the set s is the graph of an ordinal-valued function with domain η. For a predicate H, the notation $H : \mathrm{Ord} \to \mathrm{Ord}$ asserts that H consists of ordinal pairs and gives a unique value at every ordinal; $H : \eta \to \mathrm{Ord}$ has the same meaning with domain exactly η. All occurrences of Ord refer to the ordinals of the rank structure.

(a) The height is a limit ordinal greater than ω:

$$\phi_1 \equiv (\exists w)(w = \omega) \ \wedge \ (\forall \xi)(\exists \eta)\, \xi < \eta.$$

Here $w = \omega$ abbreviates the first-order definition of the least infinite ordinal.

(b) The sections $R_0, \dots, R_4$ give unique ordinal values at every appropriate tuple of ordinal indices:

$$\begin{aligned}
\phi_2 &\equiv (\exists!\eta)\, \langle 0, \eta\rangle \in R,\\
\phi_3 &\equiv (\forall \xi)(\exists!\eta)\, \langle 1, \xi, \eta\rangle \in R,\\
\phi_4 &\equiv (\forall \alpha)(\forall i)(\exists!\eta)\, \langle 2, \alpha, i, \eta\rangle \in R,\\
\phi_5 &\equiv (\forall \alpha)(\forall i)(\exists!\eta)\, \langle 3, \alpha, i, \eta\rangle \in R,\\
\phi_6 &\equiv (\forall \alpha)(\forall i)(\forall \xi)(\exists!\eta)\, \langle 4, \alpha, i, \xi, \eta\rangle \in R.
\end{aligned}$$

The five lines give the values of $\ell_p, h_p(\xi), \ell_{\alpha,i}, \delta_{\alpha,i}$, and $h_{\alpha,i}(\xi)$, respectively.

(c) The coded stems exist as sets and are normalized:

$$\begin{aligned}
\phi_7 \equiv (\exists s)\big[\, \langle 6, s\rangle \in R \ \wedge \ s : \ell_p \to \mathrm{Ord}\\
\wedge\, (\forall \xi < \ell_p)\, s(\xi) = h_p(\xi)\,\big],
\end{aligned}$$

$$\begin{aligned}
\phi_8 \equiv (\forall \alpha)(\forall i)(\exists t)\big[\, \langle 7, \alpha, i, t\rangle \in R \ \wedge \ t : \ell_{\alpha,i} \to \mathrm{Ord}\\
\wedge\, (\forall \xi < \ell_{\alpha,i})\, t(\xi) = h_{\alpha,i}(\xi)\\
\wedge\, s_p \subseteq t \ \wedge (\forall \xi)\, h_p(\xi) \le h_{\alpha,i}(\xi)\,\big].
\end{aligned}$$

The last line of ϕ_8 requires each coded condition to extend p.

(d) The ordinals coded by R_5 are unbounded:

$$\phi_9 \equiv (\forall \xi)(\exists \eta)\, [\, \xi < \eta \ \wedge \ \langle 5, \eta\rangle \in R\,].$$

(e) Predensity is expressed by the formula

$$\begin{aligned}
\phi_{10} \equiv (\forall H)\big[\, (H : \mathrm{Ord} \to \mathrm{Ord}) \implies (\forall \alpha)(\forall s)\big[\, ((\exists \eta)\, s : \eta \to \mathrm{Ord})\\
\wedge\, s \subseteq H \ \wedge \ s_p \subseteq s\\
\wedge\, (\forall \xi)\, h_p(\xi) \le H(\xi)\\
\implies (\exists i)\ \mathrm{Comp}_R(s, H, \alpha, i)\,\big]\,\big].
\end{aligned}$$

The antecedent says that (s, H) is normalized and extends p, with s a set in the rank structure. The formula $\mathrm{Comp}_R(s, H, \alpha, i)$ is the following first-order test, using the stem t supplied by R_7:

$$\begin{aligned}
(\exists t)\big[\, s_{\alpha,i} = t \ \wedge \ \big([s \subseteq t \ \wedge \ (\forall \xi < \mathrm{dom}(t))\, H(\xi) \le t(\xi)]\\
\vee\, [t \subseteq s \ \wedge \ (\forall \xi < \mathrm{dom}(s))\, h_{\alpha,i}(\xi) \le s(\xi)]\big)\,\big].
\end{aligned}$$

(f) Every ordinal-valued function on an ordinal has bounded range:

$$\phi_{11} \equiv (\forall H)(\forall \eta)\big[\, (H : \eta \to \mathrm{Ord}) \implies (\exists \zeta)(\forall \xi < \eta)\, H(\xi) < \zeta\,\big].$$

The formulas $\phi_1, \dots, \phi_9$ are first-order. In ϕ_{10} and ϕ_{11}, the assertions about function graphs are first-order once H is given. Combining the two universal second-order quantifiers gives

$$\Phi(R) = \forall H\, \Psi(R, H), \qquad \Psi \text{ first-order.} \tag{2}$$

Here R is the only fixed predicate; the height is not a parameter. The side function is a universally quantified second-order graph, while the stem is quantified as an element of the structure.

All eleven formulas hold in $(V_\kappa, \in, R)$. The formulas $\phi_1, \dots, \phi_9$ follow from the construction, ϕ_{10} from maximality of A_α below p and (1), and ϕ_{11} from regularity of κ. By Π^1_1-indescribability there is a $\beta < \kappa$ such that

$$(V_\beta, \in, R \cap V_\beta) \models \Phi(R \cap V_\beta).$$

The formula ϕ_1 makes $\beta > \omega$ a limit ordinal. At a limit height, a finite tuple belongs to V_β exactly when all its entries do. Thus ϕ_9 implies that $K' \cap \beta$ is unbounded in β. Since K' is closed, $\beta \in K' \subseteq K \cap B$, so β is an uncountable strong-limit cardinal. For every $\eta < \beta$, the graph of every function from η to β is a subset of V_β. Under full second-order semantics, ϕ_{11} says that all these functions have bounded range. Hence $\mathrm{cf}(\beta) = \beta$, and β is strongly inaccessible.

The reflected section R_0 must contain its unique original value $\ell(p)$, so $\ell(p) < \beta$. For every $\xi < \beta$, totality of R_1 similarly requires the tuple with value $h_p(\xi)$, giving $h_p(\xi) < \beta$. The reflected sections R_2, R_3, R_4 give, for all $\alpha, i < \beta$,

$$\ell(a_{\alpha,i}) < \beta, \qquad \delta_{\alpha,i} < \beta, \qquad h_{\alpha,i}[\beta] \subseteq \beta.$$

The formulas ϕ_7 and ϕ_8 supply s_p and $s_{\alpha,i}$ as elements of V_β: the stem occurs as an entry of its tuple, so a witness from the restricted predicate is the original stem. We have proved (i) and (ii).

For (iii), fix $\alpha < \beta$ and a condition $(t, H) \le p \restriction \beta$ in $\mathbb{D}_\beta$. Regularity of β bounds the range of t below β, so $t \in V_\beta$. The graph of $H : \beta \to \beta$ is a subset of V_β. The formula ϕ_{10} therefore applies and yields $i < \beta$ satisfying the compatibility test with stem $s_{\alpha,i}$. This is precisely (1) in $\mathbb{D}_\beta$, so $a_{\alpha,i} \restriction \beta$ is compatible with (t, H). This proves (iii), and $\beta \in B$ proves stationarity. □

Definition 3.9 (Local domination)**.** For a regular uncountable κ, a set $I \subseteq \mathrm{Inacc}(\kappa)$, functions $\vec{f} = \langle f_\beta : \beta \in I \rangle$ with $f_\beta : \beta \to \beta$, and a function d from κ to κ, put

$$T(\vec{f}, d) = \{\beta \in I : d[\beta] \subseteq \beta \ \wedge \ f_\beta \le^* d \restriction \beta\}.$$

Theorem 3.10. *Suppose that κ is weakly compact in V, let $I = \mathrm{Inacc}^V(\kappa)$, and let $\vec{f} = \langle f_\beta : \beta \in I \rangle \in V$, where $f_\beta : \beta \to \beta$ for each $\beta \in I$. Then*

$$\mathbb{D}_\kappa \Vdash T(\vec{f}, d) \text{ is stationary in } \kappa.$$

Proof. Work in V. By Lemma 3.7, names for ordinal sequences of length less than κ can be decided by a descending recursion with a lower bound. Thus $\mathbb{D}_\kappa$ adds no such sequences, preserves regularity of κ, and preserves the inaccessible cardinals below it.

Let $p \in \mathbb{D}_\kappa$ force that $\dot{E} \subseteq \kappa$ is club. For each $\alpha < \kappa$, every condition below p has an extension deciding an ordinal above α that belongs to $\dot{E}$, because p forces unboundedness. Choose a maximal antichain A_α below p consisting of such conditions, enumerate it as in Lemma 3.8, and choose $\delta_{\alpha,i}$ so that

$$\alpha < \delta_{\alpha,i} < \kappa, \qquad a_{\alpha,i} \Vdash \delta_{\alpha,i} \in \dot{E}.$$

Choose a strongly inaccessible β satisfying the conclusions of that lemma, and put $\ell = \ell(p)$. Replace the side function of p by

$$h^0(\xi) = \begin{cases} \max\{h_p(\xi), f_\beta(\xi)\}, & \ell \le \xi < \beta, \\ h_p(\xi), & \xi < \ell \text{ or } \beta \le \xi < \kappa. \end{cases}$$

Then $q_0 = (s_p, h^0) \le p$, and $h^0[\beta] \subseteq \beta$. In $\mathbb{D}_\beta$, each A^β_α, $\alpha < \beta$, remains predense below $q_0 \restriction \beta$.

Construct a decreasing sequence $\langle q_\alpha = (t_\alpha, k_\alpha) : \alpha < \beta \rangle$ with this initial condition, maintaining

$$\mathrm{dom}(t_\alpha) < \beta, \qquad k_\alpha[\beta] \subseteq \beta, \qquad q_\alpha \le q_0. \tag{3}$$

At stage $\alpha < \beta$, predensity gives $i_\alpha < \beta$ such that $q_\alpha \restriction \beta$ is compatible with $a_{\alpha,i_\alpha} \restriction \beta$. By (1), the corresponding global conditions are compatible as well. The longer stem and the pointwise maximum of their side functions give a common extension with stem length below β and side function mapping β into β. Extend its stem, if necessary, to length at least $\alpha + 1$, using the same side function on the new coordinates. Call the result $q_{\alpha+1}$. It satisfies (3) and extends a_{α,i_α}.

At a limit $\rho < \beta$, use the union of the previous stems and the pointwise supremum of the previous side functions. Regularity of β bounds the stem length and every value at coordinates below β by β. At coordinates at least β, regularity of κ bounds the values by κ. Once a coordinate belongs to a stem, its later side-function values equal that stem value, as in Lemma 3.7; this gives normalization. Hence the resulting condition satisfies (3).

Finally set

$$t^* = \bigcup_{\alpha<\beta} t_\alpha, \qquad k^*(\xi) = \sup_{\alpha<\beta} k_\alpha(\xi) \quad (\xi < \kappa).$$

The domain of t^* is exactly β, since every stem has length below β and $\mathrm{dom}(t_{\alpha+1}) \geq \alpha+1$. To bound its values below β, fix $\xi < \beta$ and choose $\alpha < \beta$ with $\xi < \mathrm{dom}(t_{\alpha+1})$. The value $v = t_{\alpha+1}(\xi) = k_{\alpha+1}(\xi)$ is below β; all later side functions have this value at ξ, and all earlier ones have value at most v. Consequently

$$t^*(\xi) = k^*(\xi) = v < \beta.$$

For $\xi \geq \beta$, the supremum involves only $\beta < \kappa$ values, so $k^*(\xi) < \kappa$. Therefore $q = (t^*, k^*)$ is a normalized condition in $\mathbb{D}_\kappa$ extending every q_α. It forces

$$d \restriction \beta = t^*, \qquad d[\beta] \subseteq \beta, \qquad (\forall \xi \in [\ell, \beta))\, f_\beta(\xi) \leq d(\xi).$$

Also $q \leq a_{\alpha,i_\alpha}$ for each $\alpha < \beta$, so it forces all the ordinals δ_{α,i_α} into $\dot{E}$. They are below β and cofinal in β, since $\alpha < \delta_{\alpha,i_\alpha}$. Closedness of $\dot{E}$ now gives

$$q \Vdash \beta \in \dot{E} \cap T(\vec{f}, d).$$

Since this construction applies below every condition forcing a club, $\mathbb{D}_\kappa$ forces $T(\vec{f}, d)$ to be stationary. □

Definition 3.11 (Limit closure points)**.** For a regular uncountable κ and a function d from κ to κ, put

$$E_d = \{\delta < \kappa : \delta > 0 \text{ is limit and } d[\delta] \subseteq \delta\}.$$

The pressing-down lemma makes the local domination bound constant on a stationary set of indices. A single tail D of E_d then satisfies $D \cap \beta \subseteq C_\beta$ at these indices.

Corollary 3.12. *Suppose that κ is weakly compact in V, let $I = \mathrm{Inacc}^V(\kappa)$, and let $\langle C_\beta : \beta \in I\rangle \in V$ consist of clubs $C_\beta \subseteq \beta$. If $G \subseteq \mathbb{D}_\kappa$ is generic over V and $d = d_G$, then in $V[d]$ there are a club $D \subseteq \kappa$ and a stationary set $T \subseteq I$ such that*

$$D \cap \beta \subseteq C_\beta \qquad (\beta \in T).$$

In particular, every ground-model C-sequence whose domain contains I is non-amenable in $V[d]$.

Proof. In V, for each $\beta \in I$ form the next-point function

$$n_\beta(\xi) = \min(C_\beta \setminus (\xi+1)) \qquad (\xi < \beta).$$

Unboundedness gives $\xi < n_\beta(\xi) < \beta$. Apply Theorem 3.10 to these functions. In $V[d]$, κ is regular and

$$S = \{\beta \in I : d[\beta] \subseteq \beta \ \wedge\ n_\beta \leq^* d \restriction \beta\}$$

is stationary. For $\beta \in S$, let $\gamma_\beta < \beta$ be the least bound witnessing $n_\beta \leq^* d \restriction \beta$. The pressing-down lemma, applied in $V[d]$, gives a stationary $T \subseteq S$ and an ordinal $\gamma < \kappa$ such that $\gamma_\beta = \gamma$ for all $\beta \in T$.

The set E_d is closed: if $\delta < \kappa$ is a limit of its members, then for each $\xi < \delta$ there is $\varepsilon \in E_d \cap \delta$ with $\xi < \varepsilon$, giving $d(\xi) < \varepsilon < \delta$. It is also unbounded. Above any prescribed bound, choose a strictly increasing sequence $\langle \eta_n : n < \omega\rangle$ such that $\sup(d[\eta_n]) < \eta_{n+1}$. Regularity of κ permits each choice and gives $\delta = \sup_{n<\omega} \eta_n < \kappa$; this δ belongs to E_d. Therefore $D = E_d \setminus (\gamma+1)$ is club.

Fix $\beta \in T$ and $\delta \in D \cap \beta$. Then

$$\xi < n_\beta(\xi) \leq d(\xi) < \delta \qquad (\gamma \leq \xi < \delta).$$

For every $\eta < \delta$, choose $\xi \in [\max\{\eta, \gamma\}, \delta)$; this is possible because $\delta > \gamma$ is limit. The point $n_\beta(\xi)$ lies in $C_\beta \cap \delta$ and is greater than η. Hence $C_\beta \cap \delta$ is unbounded in δ, and closedness of C_β gives $\delta \in C_\beta$. This proves the asserted inclusion. For a C-sequence with domain containing I, the resulting capture set contains T and is therefore stationary in $V[d]$. □

4. Hechler iterations

We present $< \kappa$-support iterations as partial functions whose domains are their supports. Stronger conditions are smaller in the order. Unless specified otherwise, the iteration and all cardinality calculations are computed in a fixed ground universe V. At each stage the iterand is the full normalized Hechler order of the intermediate extension. The decided-stem, Δ-system, and name-counting arguments below adapt [12, Lemma 3.10], omitting the second forcing factor used there.

Definition 4.1 (The Hechler iteration)**.** Let κ be strongly inaccessible and let λ be an ordinal. Define the $< \kappa$-support iteration

$$\langle \mathbb{P}_\alpha : \alpha \leq \lambda \rangle, \qquad \langle \dot{\mathbb{Q}}_\alpha : \alpha < \lambda \rangle$$

recursively, with $\mathbb{P}_0 = \{\varnothing\}$ and

$$\mathbf{1}_\alpha \Vdash_{\mathbb{P}_\alpha} \dot{\mathbb{Q}}_\alpha = \mathbb{D}_\kappa^{V[G_\alpha]}.$$

The greatest condition of each $\mathbb{P}_\alpha$ is the empty function, denoted by $\mathbf{1}_\alpha$. The name for the greatest condition of the iterand is $\dot{\mathbf{1}}_\alpha = (\check{\varnothing}, \mathbf{0})$, where $\mathbf{0}$ is the constant zero function from κ to κ.

Choose $\dot{\mathbb{Q}}_\alpha$ to be a *full* name: for every $r \in \mathbb{P}_\alpha$ and every $\mathbb{P}_\alpha$-name τ,

$$r \Vdash \tau \in \dot{\mathbb{Q}}_\alpha \quad \Longrightarrow \quad (\exists \sigma \in \mathrm{dom}(\dot{\mathbb{Q}}_\alpha))\, r \Vdash \sigma = \tau. \tag{4}$$

Here $\mathrm{dom}(\dot{\mathbb{Q}}_\alpha)$ is the set of names occurring as first entries in $\dot{\mathbb{Q}}_\alpha$. Such a name can be chosen with a set-sized domain. To see this, mix a name forced by r to be a Hechler condition with $\dot{\mathbf{1}}_\alpha$ on the other members of a maximal antichain containing r. The mixed name is globally forced to be a Hechler condition. Its two component graphs have names whose entries are check names for pairs in $\kappa \times \kappa$, paired with conditions of $\mathbb{P}_\alpha$. These graph names form a set. Take all ordered-pair names formed from them that are globally forced to be normalized Hechler conditions, each paired with $\mathbf{1}_\alpha$, and include $\dot{\mathbf{1}}_\alpha$. Every Hechler condition in an extension has such a representative, and the mixed representative agrees with the original name below r. This gives (4) without further strengthening r.

For $0 < \theta \leq \lambda$, a condition $p \in \mathbb{P}_\theta$ is a partial function with

$$\mathrm{supp}(p) = \mathrm{dom}(p) \subseteq \theta, \qquad |\mathrm{supp}(p)| < \kappa,$$

such that $p \restriction \alpha \in \mathbb{P}_\alpha$ for every $\alpha < \theta$, and, at each $\alpha \in \mathrm{dom}(p)$,

$$p(\alpha) \in \mathrm{dom}(\dot{\mathbb{Q}}_\alpha), \qquad p \restriction \alpha \Vdash_{\mathbb{P}_\alpha} p(\alpha) \in \dot{\mathbb{Q}}_\alpha.$$

Membership in the iterand is required below $p \restriction \alpha$. Whenever $\alpha \notin \mathrm{dom}(p)$, the expression $p(\alpha)$ is understood to mean $\dot{\mathbf{1}}_\alpha$; this convention does not add α to the domain. Initial segments are included by leaving all later coordinates omitted. At a limit stage we take all such partial functions whose restrictions belong to the earlier stages, including unbounded supports when the cofinality of the stage is less than κ.

For $p, q \in \mathbb{P}_\theta$, the order is

$$\begin{aligned} q \leq p \quad \Longleftrightarrow \quad & \mathrm{supp}(p) \subseteq \mathrm{supp}(q) \ \text{ and} \\ & (\forall \alpha \in \mathrm{supp}(p))\, q \restriction \alpha \Vdash_{\mathbb{P}_\alpha} q(\alpha) \leq_{\dot{\mathbb{Q}}_\alpha} p(\alpha). \end{aligned} \tag{5}$$

At a coordinate added by q, validity of q already ensures that its value extends the greatest condition assigned to that coordinate in p. A coordinate in $\mathrm{dom}(p)$ remains in the support even when its value is forced to be the greatest condition. Thus one may insert coordinates with trivial values, and strengthening retains every coordinate in the domain. These presentations are preorders; identifying mutually extending conditions gives partial orders.

Definition 4.2 (Decided stems)**.** For $p \in \mathbb{P}_\lambda$ and $\alpha < \lambda$, choose $\mathbb{P}_\alpha$-names $\dot{s}_\alpha^p$ and $\dot{h}_\alpha^p$ with

$$p \restriction \alpha \Vdash p(\alpha) = \langle \dot{s}_\alpha^p, \dot{h}_\alpha^p \rangle.$$

Below this initial condition, the names denote a stem in ${}^{<\kappa}\kappa$ and a function in ${}^{\kappa}\kappa$, respectively. At omitted coordinates use the empty stem and the zero function. Any two choices of component names agree below $p \restriction \alpha$, so the following definition is independent of those choices. Define

$$\mathcal{E}_\lambda = \{p \in \mathbb{P}_\lambda : (\forall \alpha \in \mathrm{supp}(p))(\exists s_\alpha \in {}^{<\kappa}\kappa)\, p \restriction \alpha \Vdash \dot{s}_\alpha^p = \check{s}_\alpha\}.$$

Members of $\mathcal{E}_\lambda$ have *decided stems*; their side-function names remain arbitrary.

Lemma 4.3. *Let $\mathbb{P}_\lambda$ be the Hechler iteration at a strongly inaccessible κ, and let $\theta \leq \lambda$. The inclusion $\mathbb{P}_\theta \subseteq \mathbb{P}_\lambda$ preserves incompatibility and sends maximal antichains to predense sets. Consequently, if G_λ is generic over V, then $G_\theta = G_\lambda \cap \mathbb{P}_\theta$ is $\mathbb{P}_\theta$-generic over V. If $\mathbb{P}_\theta$ preserves κ and adds no ordinal sequences of length less than κ, then, in $V[G_\theta]$, the remaining forcing is forcing equivalent to the full $< \kappa$-support Hechler iteration on $[\theta, \lambda)$, computed in that universe.*

Proof. The case $\theta = \lambda$ is immediate. Suppose $\theta < \lambda$. Initial-segment completeness and genericity are the standard iteration facts recalled in [5, Section 7, pp. 799–800]. In our partial-function presentation, the restriction map is a projection: if $p \in \mathbb{P}_\lambda$ and $r \leq p \restriction \theta$, then

$$r \frown_\theta p = r \cup (p \restriction [\theta, \lambda)) \tag{6}$$

is a condition below both p and r.

By [5, Proposition 7.11], the quotient is forcing equivalent to the factor iteration obtained by translating the remaining iterand names into $V[G_\theta]$. Fullness, (4), permits choosing coordinate representatives in our iterand-name domains without further strengthening. At successor stages the translated iterands are the full Hechler orders of the corresponding intermediate extensions.

For the limit stages, the hypothesis gives

$$([\lambda]^{<\kappa})^{V[G_\theta]} = ([\lambda]^{<\kappa})^V,$$

since an enumeration of a set on the left is an ordinal sequence of length less than κ. More generally, there are no new sequences of fewer than κ ground-model objects. Thus the limit-stage construction of the factor iteration admits exactly the $< \kappa$-supports of the full tail, including unbounded supports at stages of cofinality less than κ. Consequently,

$$\mathbb{P}_\lambda \simeq \mathbb{P}_\theta * \dot{\mathbb{P}}_{[\theta,\lambda)},$$

where the second factor is the full $< \kappa$-support Hechler iteration computed in $V[G_\theta]$. □

Lemma 4.4. *Let κ be strongly inaccessible. For every ordinal λ, the Hechler iteration $\mathbb{P}_\lambda$ is $< \kappa$-closed and κ^+-Knaster, and $\mathcal{E}_\lambda$ is dense in $\mathbb{P}_\lambda$. Moreover, for every $\theta \leq \lambda$,*

$$\mathbb{P}_\theta \Vdash \textit{the tail iteration } \mathbb{P}_{[\theta,\lambda)} \textit{ is } <\kappa\textit{-closed},$$

where the tail is computed in the initial generic extension as in Lemma 4.3.

Proof. Closure and tails. We prove closure and preservation of strong inaccessibility simultaneously by induction on λ. Let $\langle p_i : i < \delta \rangle$ be a decreasing sequence in $\mathbb{P}_\lambda$, where $0 < \delta < \kappa$, and put

$$u = \bigcup_{i<\delta} \operatorname{supp}(p_i), \qquad |u| < \kappa.$$

Construct q recursively on u. Suppose that $r_\alpha = q \restriction \alpha \leq p_i \restriction \alpha$ for every $i < \delta$. Choose $\mathbb{P}_\alpha$-names $\dot{t}_\alpha$ and $\dot{g}_\alpha$ such that

$$r_\alpha \Vdash \dot{t}_\alpha = \bigcup_{i<\delta} \dot{s}_\alpha^{p_i}, \qquad r_\alpha \Vdash \dot{g}_\alpha = \left\langle \sup_{i<\delta} \dot{h}_\alpha^{p_i}(\xi) : \xi < \kappa \right\rangle.$$

Choose a name $\dot{b}_\alpha$ with

$$r_\alpha \Vdash \dot{b}_\alpha = \langle \dot{t}_\alpha, \dot{g}_\alpha \rangle.$$

By the induction hypothesis κ remains regular after $\mathbb{P}_\alpha$. Under r_α, the coordinate values form a decreasing sequence of Hechler conditions, with the maximum used at omitted coordinates. Lemma 3.7 gives

$$r_\alpha \Vdash \dot{b}_\alpha \in \dot{\mathbb{Q}}_\alpha \ \wedge\ (\forall i < \delta)\, \dot{b}_\alpha \leq p_i(\alpha).$$

By (4), choose $q(\alpha) \in \operatorname{dom}(\dot{\mathbb{Q}}_\alpha)$ with $r_\alpha \Vdash q(\alpha) = \dot{b}_\alpha$. Set $\operatorname{dom}(q) = u$, omitting all coordinates outside u. Induction using (5) yields $q \in \mathbb{P}_\lambda$ and $q \leq p_i$ for all $i < \delta$. Denote this lower bound by $\mathrm{lb}_\lambda(\langle p_i : i < \delta \rangle)$. Its support is exactly u, including coordinates whose lower-bound value is trivial. At each coordinate its initial restriction forces its stem to be the union of the corresponding stems. For an empty sequence take $\mathbf{1}_\lambda$.

For a name $\dot{f} : \eta \to \mathrm{Ord}$, with $\eta < \kappa$, and a condition forcing its stated domain, construct $\langle r_\xi : \xi \leq \eta \rangle$ below that condition with

$$r_{\xi+1} \Vdash \dot{f}(\xi) = \check{\beta}_\xi, \qquad r_\zeta = \mathrm{lb}_\lambda(\langle r_\xi : \xi < \zeta \rangle) \quad (0 < \zeta \leq \eta \text{ limit}).$$

Then $r_\eta \Vdash \dot f = \langle \beta_\xi : \check\xi < \eta\rangle$. Thus no ordinal sequences of length less than κ are added. In particular, no subsets of ordinals below κ are added, no cardinal below κ is collapsed, and κ remains regular. It therefore remains strongly inaccessible, completing the induction. Lemma 4.3 identifies the quotient after $\mathbb{P}_\theta$ with the full Hechler tail in $V[G_\theta]$; the same lower-bound recursion in that universe proves its $<\kappa$-closure.

Density of decided stems. Fix $p \in \mathbb{P}_\lambda$. Starting with $p_0 = p$, construct a decreasing sequence $\langle p_n : n < \omega\rangle$. At stage n, enumerate

$$\operatorname{supp}(p_n) = \{\alpha_{n,\xi} : \xi < \mu_n\}, \qquad \mu_n < \kappa,$$

and build decreasing conditions $\langle p_{n,\xi} : \xi \le \mu_n\rangle$, starting with $p_{n,0} = p_n$. At $\xi < \mu_n$, let $\alpha = \alpha_{n,\xi}$. First strengthen $p_{n,\xi} \restriction \alpha$ to decide the stem length. Then use $<\kappa$-closure of $\mathbb{P}_\alpha$ to decide all values of that stem. This gives $v_{n,\xi} \le p_{n,\xi} \restriction \alpha$ and $t_{\alpha,n} \in {}^{<\kappa}\kappa$ such that

$$v_{n,\xi} \Vdash \dot s_\alpha^{p_{n,\xi}} = \check t_{\alpha,n}.$$

Set

$$p_{n,\xi+1} = v_{n,\xi} \cup (p_{n,\xi} \restriction [\alpha,\lambda)).$$

By (6), this is a condition below $p_{n,\xi}$, and its initial restriction forces its stem at α to be $t_{\alpha,n}$. At a nonzero limit $\zeta \le \mu_n$, use the specified lower bound

$$p_{n,\zeta} = \mathrm{lb}_\lambda(\langle p_{n,\xi} : \xi < \zeta\rangle).$$

Finally put $p_{n+1} = p_{n,\mu_n}$. Supports increase throughout the construction, and the lower-bound operation takes their union. Hence every coordinate introduced during stage n belongs to $\operatorname{supp}(p_{n+1})$ and is treated at every later stage. Each stage involves fewer than κ steps, so all its conditions have support of size less than κ.

Take

$$q = \mathrm{lb}_\lambda(\langle p_n : n < \omega\rangle), \qquad \operatorname{supp}(q) = \bigcup_{n<\omega} \operatorname{supp}(p_n).$$

Fix $\alpha \in \operatorname{supp}(q)$, and choose n_α so that $\alpha \in \operatorname{supp}(p_n)$ for all $n \ge n_\alpha$. The stem decision at stage n, together with the decreasing order, gives

$$q \restriction \alpha \Vdash \dot s_\alpha^{p_n} \subseteq \check t_{\alpha,n} \subseteq \dot s_\alpha^{p_{n+1}} \quad (n \ge n_\alpha).$$

In particular, $t_{\alpha,n} \subseteq t_{\alpha,n+1}$. Regularity of κ gives $t_\alpha = \bigcup_{n\ge n_\alpha} t_{\alpha,n} \in {}^{<\kappa}\kappa$. The two inclusions above show that the union of the stems of the p_n's is exactly t_α; stems from earlier stages are contained in t_{α,n_α}. By the choice of lower bound,

$$q \restriction \alpha \Vdash \dot s_\alpha^{q} = \bigcup_{n<\omega} \dot s_\alpha^{p_n} = \check t_\alpha.$$

Thus $q \in \mathcal{E}_\lambda$ and $q \le p$.

The Knaster property. Given $\langle p_i : i < \kappa^+\rangle$, choose $q_i \in \mathcal{E}_\lambda$ with $q_i \le p_i$, and put $u_i = \operatorname{supp}(q_i)$. Since $\kappa^{<\kappa} = \kappa$, the generalized Δ-system lemma gives $J \subseteq \kappa^+$ and a root R with

$$|J| = \kappa^+, \qquad |R| < \kappa, \qquad u_i \cap u_j = R \quad (i \ne j \text{ in } J).$$

If one support occurs κ^+ times, use it as the root; otherwise there are κ^+ distinct supports to which the lemma applies. The number of assignments of decided stems on R is

$$\left|{}^R({}^{<\kappa}\kappa)\right| = \kappa^{|R|} \le \kappa.$$

Thus there is $J' \subseteq J$ of size κ^+ and stems s_α, $\alpha \in R$, with

$$q_i \restriction \alpha \Vdash \dot s_\alpha^{q_i} = \check s_\alpha \quad (i \in J',\ \alpha \in R).$$

For distinct $i, j \in J'$, construct w recursively on $u_i \cup u_j$, retaining all these coordinates. Off the root use the name from the unique condition whose support contains the coordinate; omit coordinates outside $u_i \cup u_j$. Inductively, suppose $w \restriction \alpha \le q_i \restriction \alpha, q_j \restriction \alpha$. At a root coordinate choose names $\dot g_\alpha$ and $\dot c_\alpha$ such that

$$w \restriction \alpha \Vdash \dot g_\alpha = \left\langle \max\{\dot h_\alpha^{q_i}(\xi), \dot h_\alpha^{q_j}(\xi)\} : \xi < \kappa \right\rangle,$$

$$w \restriction \alpha \Vdash \dot c_\alpha = \langle \check s_\alpha, \dot g_\alpha\rangle.$$

This initial restriction forces both stems to be s_α. Normalization implies that both side functions agree with s_α on its domain. Their pointwise maximum therefore does as well, and

$$w \restriction \alpha \Vdash \dot c_\alpha \in \dot{\mathbb{Q}}_\alpha \ \wedge\ \dot c_\alpha \le q_i(\alpha), q_j(\alpha).$$

Fullness gives $w(\alpha) \in \mathrm{dom}(\dot{\mathbb{Q}}_\alpha)$ with $w \restriction \alpha \Vdash w(\alpha) = \dot{c}_\alpha$. Outside the root, the retained name is valid below the stronger initial restriction and extends the greatest condition assigned to that coordinate in the other condition. Since $|u_i \cup u_j| < \kappa$, the recursion defines $w \in \mathbb{P}_\lambda$ with $w \leq q_i, q_j$, and hence $w \leq p_i, p_j$. This proves κ^+-Knaster.

To record the cardinal preservation needed below, let $\dot{f} : \eta \to \kappa^+$ be a name, with $\eta \leq \kappa$. Choose maximal antichains deciding its values and let X_ξ be the set of possible values at $\xi < \eta$. Then

$$|X_\xi| \leq \kappa, \qquad \left|\bigcup_{\xi<\eta} X_\xi\right| \leq \kappa, \qquad \Vdash \mathrm{ran}(\dot{f}) \subseteq \bigcup_{\xi<\eta} X_\xi.$$

The union is bounded in the ground-model κ^+, so no such map is cofinal. Since κ is preserved, the ground-model κ^+ remains its regular successor. □

Lemma 4.5. *Let κ be strongly inaccessible and put $\mu = 2^\kappa$. For every $\theta \leq \kappa^+$, the Hechler iteration $\mathbb{P}_\theta$ has a $< \kappa$-closed dense suborder of size at most μ.*

Proof. All counts are in V. Cardinal arithmetic gives

$$\mu^\kappa = \mu, \qquad \mu^{<\kappa} = \mu, \qquad \kappa^+ \leq \mu.$$

Construct dense sets $D_\theta \subseteq \mathbb{P}_\theta$ recursively, starting with $D_0 = \mathbb{P}_0$. Suppose D_α has been constructed for each $\alpha < \theta$ and $|D_\alpha| \leq \mu$.

For $\alpha < \theta$, let H_α be the set of names for graphs of functions of the form

$$\dot{h}_{\vec{A},\vec{\nu}} = \{(\langle \xi, \check{\nu_\xi}(a)\rangle, a) : \xi < \kappa,\ a \in A_\xi\},$$

where each $A_\xi \subseteq D_\alpha$ is a maximal antichain of $\mathbb{P}_\alpha$ and $\nu_\xi : A_\xi \to \kappa$. Such a name is globally forced to be a function from κ to κ. Conversely, for every globally total function name $\dot{h}$, density of D_α gives maximal antichains $A_\xi \subseteq D_\alpha$ and assignments with

$$a \Vdash \dot{h}(\xi) = \check{\nu_\xi}(a) \quad (\xi < \kappa,\ a \in A_\xi).$$

The corresponding member of H_α is globally forced equal to $\dot{h}$. By the chain condition, each A_ξ has size at most κ. Thus each presentation is coded by at most κ triples in $\kappa \times D_\alpha \times \kappa$, and

$$|H_\alpha| \leq (\kappa \cdot \mu \cdot \kappa)^\kappa = \mu.$$

For $s \in {}^{<\kappa}\kappa$ and $\dot{h} \in H_\alpha$, define the normalized function name $\dot{h}[s]$ by

$$\mathbf{1}_\alpha \Vdash \dot{h}[s](\xi) = \begin{cases} s(\xi), & \xi < \mathrm{dom}(s), \\ \dot{h}(\xi), & \mathrm{dom}(s) \leq \xi < \kappa. \end{cases}$$

Then $\mathbf{1}_\alpha \Vdash \langle \check{s}, \dot{h}[s]\rangle \in \dot{\mathbb{Q}}_\alpha$. For each such $s, \dot{h}$, fullness gives a name $\sigma \in \mathrm{dom}(\dot{\mathbb{Q}}_\alpha)$; fix one with

$$\mathbf{1}_\alpha \Vdash \sigma = \langle \check{s}, \dot{h}[s]\rangle.$$

Let F_α consist of these chosen names together with $\dot{\mathbf{1}}_\alpha$. Since $|{}^{<\kappa}\kappa| = \kappa$, $|F_\alpha| \leq \mu$. Define

$$D_\theta = \{p \in \mathbb{P}_\theta : (\forall \alpha \in \mathrm{supp}(p))\ p(\alpha) \in F_\alpha\}.$$

To prove density, fix $p \in \mathbb{P}_\theta$ and choose $p' \in \mathcal{E}_\theta$ with $p' \leq p$. For $\alpha \in \mathrm{supp}(p')$, let s_α be its decided stem. The component name $\dot{h}^{p'}_\alpha$ is forced to be a function only below $p' \restriction \alpha$. First extend it to a globally total function-name by mixing with the zero function off that cone: use a maximal antichain containing $p' \restriction \alpha$, retain the original name there, and use zero on the other members. Then choose its nice representative $\dot{h}_\alpha \in H_\alpha$. Thus

$$p' \restriction \alpha \Vdash \dot{h}_\alpha = \dot{h}^{p'}_\alpha.$$

Normalization of $p'(\alpha)$ gives

$$p' \restriction \alpha \Vdash p'(\alpha) = \langle \check{s}_\alpha, \dot{h}_\alpha[s_\alpha]\rangle.$$

At every $\alpha \in \mathrm{supp}(p')$, use for $q(\alpha)$ the name in F_α chosen for $s_\alpha, \dot{h}_\alpha$, retaining that coordinate even if its value is trivial. Omit coordinates outside $\mathrm{supp}(p')$. Thus $\mathrm{supp}(q) = \mathrm{supp}(p')$. Induction on the coordinates gives

$$q \restriction \alpha \leq p' \restriction \alpha, \qquad q \restriction \alpha \Vdash q(\alpha) = p'(\alpha).$$

Hence $q \in D_\theta$ and $q \leq p' \leq p$. Finally,

$$|[\theta]^{<\kappa}| \leq \mu^{<\kappa} = \mu, \qquad |D_\theta| \leq \mu \cdot \mu^{<\kappa} = \mu.$$

This completes the recursion at successor and limit lengths. For a descending sequence of length less than κ in D_θ, Lemma 4.4 gives a lower bound in $\mathbb{P}_\theta$. Refine that lower bound into D_θ by density. Thus D_θ, with the inherited order, is $< \kappa$-closed. □

Lemma 4.6. *Let κ be strongly inaccessible, let $\lambda = \kappa^+$, and let $G_\lambda \subseteq \mathbb{P}_\lambda$ be generic over V for the Hechler iteration. Put $G_\theta = G_\lambda \cap \mathbb{P}_\theta$ for $\theta \leq \lambda$. Every subset of κ in $V[G_\lambda]$ belongs to $V[G_\theta]$ for some $\theta < \lambda$. The same holds for every C-sequence on κ.*

Proof. Let $X \subseteq \kappa$ belong to $V[G_\lambda]$. Choose a name $\dot{X}$ that is globally forced to be a subset of κ and evaluates to X. Such a name is obtained by intersecting any name for X with $\check{\kappa}$. For each $\xi < \kappa$, choose a maximal antichain B_ξ deciding whether $\check{\xi} \in \dot{X}$, and set

$$B_\xi^+ = \{p \in B_\xi : p \Vdash \check{\xi} \in \dot{X}\}, \qquad \dot{X}^* = \{(\check{\xi}, p) : \xi < \kappa,\ p \in B_\xi^+\}.$$

Then $\mathbf{1}_\lambda \Vdash \dot{X}^* = \dot{X}$. By Lemma 4.4, $|B_\xi| \leq \kappa$. The union

$$a = \bigcup_{\xi<\kappa} \bigcup_{p\in B_\xi} \operatorname{supp}(p)$$

has size at most κ. Regularity of $\lambda = \kappa^+$ gives

$$\theta = \sup\{\alpha + 1 : \alpha \in a\} < \lambda.$$

Every condition used in $\dot{X}^*$ has support below θ and is a condition of $\mathbb{P}_\theta$: at each coordinate α, its entry is already a $\mathbb{P}_\alpha$-name. Hence $\dot{X}^*$ is a $\mathbb{P}_\theta$-name and

$$X = (\dot{X}^*)^{G_\lambda} = (\dot{X}^*)^{G_\theta} \in V[G_\theta].$$

Fix in V a bijection $b : \kappa \times \kappa \to \kappa$. For a C-sequence $\vec{C}$ in the final extension, apply the first part to

$$X_{\vec{C}} = \{b(\beta, \xi) : \beta < \kappa,\ \xi \in C_\beta\}.$$

In the resulting intermediate extension, define

$$C_\beta = \{\xi < \beta : b(\beta, \xi) \in X_{\vec{C}}\} \qquad (\beta < \kappa).$$

These are the original entries. For a fixed set of ordinals, closedness and unboundedness in a fixed ordinal are absolute between the two extensions. Thus the entire sequence is already a C-sequence in $V[G_\theta]$. □

Lemma 4.7. *Let κ be regular uncountable, let $S \subseteq \kappa$ be stationary, and let $\mathbb{R}$ be $< \kappa$-closed. Then $\mathbb{R}$ preserves the stationarity of S.*

Proof. Closure adds no ordinal sequences of length $< \kappa$ and hence preserves regularity of κ. Fix $p \in \mathbb{R}$ with $p \Vdash \dot{E} \subseteq \kappa$ is club. In the ground universe construct decreasing conditions $\langle p_i : i < \kappa\rangle$, a strictly increasing continuous sequence $\langle \eta_i : i < \kappa\rangle$, and ordinals $\xi_i < \kappa$, starting with $p_0 = p$ and $\eta_0 = 0$. At successors, forced unboundedness lets us choose

$$p_{i+1} \leq p_i, \qquad \eta_i < \xi_i < \kappa, \qquad p_{i+1} \Vdash \check{\xi}_i \in \dot{E}, \qquad \eta_{i+1} = \xi_i + 1.$$

At a nonzero limit $\rho < \kappa$, choose a lower bound p_ρ of the preceding conditions and put $\eta_\rho = \sup_{i<\rho} \eta_i < \kappa$. Continuity and strict increase make

$$F = \{\eta_\rho : 0 < \rho < \kappa \text{ is a limit ordinal}\}$$

a club in κ. Choose a limit ρ with $\eta_\rho \in S$. Since p_ρ extends every p_{i+1} for $i < \rho$,

$$p_\rho \Vdash \{\xi_i : \check{i} < \rho\} \subseteq \dot{E}, \qquad \sup_{i<\rho} \xi_i = \sup_{i<\rho} \eta_{i+1} = \eta_\rho.$$

The forced closedness of $\dot{E}$ gives $p_\rho \Vdash \check{\eta}_\rho \in \dot{E} \cap \check{S}$. Repeating the construction below any extension of p shows that conditions forcing an intersection with S are dense below p, as required. □

5. Preservation and the consistency result

5.1. **Preservation under the Π^2_1 hypothesis.** For the Π^2_1 construction, choose the local Menas function f at height $\kappa+1$ specified in Proposition A.3. Let $\mathbb{R}$ be the local lottery preparation of [9], relative to f. This is the Easton support iteration of length κ which, at eligible stages $\gamma<\kappa$, chooses from the lottery sum of $<\gamma$-closed orders in $H_{f(\gamma)^+}$, calculated in the intermediate universe. Eligibility requires $\gamma\in\mathrm{dom}(f)$ and $f''\gamma\subseteq\gamma$. We use the size-κ presentation with conditions in V_κ described in [9, pp. 1229–1230]. It is κ-cc and preserves inaccessibility. By [9, Corollary 38], with $m=1$, it also preserves Π^2_1-indescribability. The indestructibility bound at this height is $\beth_\kappa=\kappa$.

Theorem 5.1. *Suppose κ is Π^2_1-indescribable and $2^\kappa=\kappa^+$. Let $G\subseteq\mathbb{R}$ be generic for the local lottery preparation above and put $U=V[G]$. Then $U\models 2^\kappa=\kappa^+$, and, for every $\theta\leq\kappa^+$, the full $<\kappa$-support Hechler iteration computed in U satisfies*

$$\mathbb{P}_\theta \Vdash_U \kappa \text{ is } \Pi^2_1\text{-indescribable.}$$

The proof is deferred to Appendix A. It uses the canonical presentation of Subsection A.2 to extend [9, Corollary 38], with $m=1$, to the full Hechler iteration.

Corollary 5.2. *In the universe prepared in Theorem 5.1, all initial segments and the full Hechler iteration of length κ^+ preserve Π^2_1-indescribability and, in particular, weak compactness.*

Proof. Apply Theorem 5.1 to each initial segment computed in the prepared universe. □

5.2. **Preservation from a Π^3_1 hypothesis.** The stronger hypothesis of Π^3_1-indescribability allows the iteration to be treated directly by [9, Corollary 38]. This application requires the local lottery preparation at height $\kappa+2$, rather than the preparation at height $\kappa+1$ used in Theorem 5.1.

Proposition 5.3. *Suppose that κ is Π^3_1-indescribable and $2^\kappa=\kappa^+$. Choose a partial local Menas function for $(\kappa+2)$-strong unfoldability, and let $\mathbb{R}_3$ be the corresponding local lottery preparation. If $G\subseteq\mathbb{R}_3$ is generic and $U=V[G]$, then $U\models 2^\kappa=\kappa^+$, and, for every $\theta\leq\kappa^+$, the full Hechler iteration computed in U satisfies*

$$\mathbb{P}_\theta \Vdash_U \kappa \text{ is } \Pi^3_1\text{-indescribable.}$$

Proof. By [9, Corollary 35], with $m=2$, the initial hypothesis is equivalent to $(\kappa+2)$-strong unfoldability. The local Menas function is supplied by [9, Theorem 10(2)] at that height. Use the size-κ, κ-cc presentation of its lottery preparation described in [9, pp. 1229–1230]. By [9, Corollary 38], again with $m=2$, the preparation preserves Π^3_1-indescribability and makes it indestructible by $<\kappa$-closed, κ-proper forcing of size at most $\beth_{\kappa+1}$.

The preparation preserves κ^+, since it is κ-cc. Every name for a subset of κ has a nice representative consisting of at most κ pairs of ordinals below κ and preparation conditions. There are at most $(2^\kappa)^\kappa=\kappa^+$ such names in the ground universe. The old subsets of κ remain distinct, so $U\models 2^\kappa=\kappa^+$. Inaccessibility in U gives $\beth_\kappa=\kappa$, and hence the bound in Corollary 38 is $\beth_{\kappa+1}=2^\kappa=\kappa^+$, computed in U.

Fix $\theta\leq\kappa^+$. In U, Lemma 4.5 gives a $<\kappa$-closed dense presentation of $\mathbb{P}_\theta$ of size at most κ^+. It is κ^+-cc by Lemma 4.4: compatibility between its conditions agrees with compatibility in the full iteration, because the presentation is dense. Since $\kappa^{<\kappa}=\kappa$, [9, Fact 13(4)] makes it κ-proper. Corollary 38 therefore applies to this presentation. Forcing equivalence gives the stated conclusion for $\mathbb{P}_\theta$. □

5.3. **Eliminating amenable C-sequences.** To eliminate amenable C-sequences, we apply the capture theorem in an intermediate universe containing the given sequence, then preserve its stationary capture set through the tail. The following theorem requires weak compactness at each proper initial stage, independently of how it is preserved.

Theorem 5.4. *Let κ be weakly compact and let $\lambda=\kappa^+$. Form the Hechler iteration $\mathbb{P}_\lambda$ of Definition 4.1, and suppose that*

$$\mathbb{P}_\theta \Vdash \kappa \text{ is weakly compact} \qquad (\theta<\lambda).$$

Then $\mathbb{P}_\lambda\Vdash\kappa\notin\mathcal{A}_\kappa$.

Proof. Let $G_\lambda\subseteq\mathbb{P}_\lambda$ be generic over the ground universe V, and let $G_\theta=G_\lambda\cap\mathbb{P}_\theta$ for $\theta\leq\lambda$. Fix a C-sequence $\vec{C}=\langle C_\beta:\beta<\kappa\rangle$ in $V[G_\lambda]$. By Lemma 4.6, there is $\theta<\lambda$ such that $\vec{C}\in V[G_\theta]$, and it is already a C-sequence there. Put $M=V[G_\theta]$. The hypothesis makes κ weakly compact in M.

Apply Corollary 3.12 inside M to the restriction of $\vec{C}$ to $\mathrm{Inacc}^M(\kappa)$. At coordinate θ, the iterand is the full order $\mathbb{D}_\kappa^M$. If d_θ is its generic function, then $M[d_\theta] = V[G_{\theta+1}]$ contains a club $D \subseteq \kappa$ and a stationary set $T \subseteq \mathrm{Inacc}^M(\kappa)$ satisfying

$$D \cap \beta \subseteq C_\beta \qquad (\beta \in T).$$

Since $\theta + 1 < \lambda$, the tail forcing from that stage to λ is $< \kappa$-closed in $V[G_{\theta+1}]$ by Lemma 4.4, and it preserves the stationarity of T by Lemma 4.7. Closedness and unboundedness of D, and the displayed inclusions, are absolute to the final extension: they concern fixed sets of ordinals. Thus $\mathrm{Cap}(D, \vec{C})$ contains the still stationary set T in $V[G_\lambda]$, so $\vec{C}$ is not amenable. Since the sequence was arbitrary, $\kappa \notin \mathcal{A}_\kappa$ in the final extension. □

Proof of Theorem 1.3. By Corollary 2.3, begin in a universe $W \models \mathsf{ZFC} + V = L$ containing a Π^2_1-indescribable cardinal. Fix the least such cardinal κ in W. Proposition 2.2, applied inside W, shows that κ is not ineffable in $W = L^W$. Apply Theorem 5.1 inside W, and let $U = W[H]$ be the resulting prepared universe.

In U, form the $< \kappa$-support Hechler iteration of length $\lambda = (\kappa^+)^U$, and let $G_\lambda \subseteq \mathbb{P}_\lambda$ be generic over U. For $\theta \leq \lambda$, let $G_\theta = G_\lambda \cap \mathbb{P}_\theta$. Corollary 5.2 gives

$$U[G_\theta] \models \kappa \text{ is } \Pi^2_1\text{-indescribable} \qquad (\theta \leq \lambda).$$

In particular, weak compactness holds at every proper initial stage, so Theorem 5.4 applies over U. The final universe $N = U[G_\lambda]$ therefore satisfies

$$N \models \kappa \text{ is } \Pi^2_1\text{-indescribable and } \kappa \notin \mathcal{A}_\kappa.$$

Set forcing leaves the ordinals and the constructible universe unchanged. Hence $L^N = L^W = W$, since $W \models V = L$. The choice of κ in W gives

$$L^N \models \kappa \text{ is not ineffable.}$$

The forcing theorem yields the asserted relative consistency implication. □

Corollary 5.5. *If* $\mathsf{ZFC}+$ *"there is a* Π^3_1*-indescribable cardinal" is consistent, then so is*

$$\mathsf{ZFC} + \exists \kappa \left[\kappa \textit{ is } \Pi^3_1\textit{-indescribable } \wedge\ \kappa \notin \mathcal{A}_\kappa\ \wedge\ L \models \kappa \textit{ is not ineffable}\right].$$

Proof. As for the Π^2_1 hypothesis, the consistency of Π^3_1-indescribability with $V = L$ is recorded in [9, Section 1, p. 1215]. Begin in $W \models \mathsf{ZFC} + V = L$ and take its least Π^3_1-indescribable cardinal κ. If κ were subtle in W, [7, Lemma 5.4] would give a totally indescribable cardinal below it, contradicting its minimality. Thus κ is not ineffable in W, and GCH supplies $2^\kappa = \kappa^+$.

Apply Proposition 5.3 in W, and then force with the full Hechler iteration of length κ^+ in the prepared universe U. The proposition preserves Π^3_1-indescribability at every initial stage, including the final stage. In particular, weak compactness holds at every proper initial stage, so Theorem 5.4 gives $\kappa \notin \mathcal{A}_\kappa$ in the final universe N. Since $W \models V = L$ and both forcings are set forcings, $L^N = W$. Consequently, L^N still regards κ as non-ineffable. □

Appendix A. Preservation under the Π^2_1 hypothesis

We prove Theorem 5.1 by adapting the two-step lifting argument of [9, pp. 1229–1232] to the full Hechler iteration. The canonical presentation below lets us identify the iteration computed by the target with the actual order in the prepared universe.

A.1. **Embedding data.**

Definition A.1 (κ-models)**.** Throughout the article, $\mathsf{ZFC}^-_{\mathrm{Coll}}$ denotes ZFC without Power Set, with full Separation, Replacement, and Collection, and with the assertion that every set can be well ordered. For inaccessible κ, a *κ-model* is a transitive set M such that

$$|M| = \kappa, \qquad \kappa \in M, \qquad M \models \mathsf{ZFC}^-_{\mathrm{Coll}}, \qquad {}^{<\kappa}M \subseteq M.$$

Size and closure are evaluated in the ambient universe.

Collection is specified because, without Power Set, it need not follow from Replacement [6].

For inaccessible κ, Π^2_1-indescribability is equivalent to the following property: for every κ-model M, there are a transitive set N and an elementary embedding $j : M \to N$ such that

$$\mathrm{crit}(j) = \kappa, \qquad j(\kappa) > \kappa + 1, \qquad V_{\kappa+1} \subseteq N.$$

Here the rank segment belongs to the ambient universe. This is $(\kappa{+}1)$-strong unfoldability in [9, Definition 3 and Corollary 35].

The characterization remains valid when κ-models are required to satisfy Collection. Every $A \subseteq \kappa$ can be placed into a model of Definition A.1 by collapsing a $< \kappa$-closed elementary submodel of a sufficiently large H_χ, of size κ, containing A and all ordinals at most κ. For regular χ, H_χ satisfies Collection. The embedding property for these models implies the subset-of-κ characterization in [9, Fact 4(4)]. Conversely, every model of Definition A.1 satisfies the κ-model requirements of [9], so the same fact supplies the embedding for each of these models. Together with [9, Corollary 35], with $m = 1$, this proves the asserted equivalence.

Definition A.2 (Local Menas property)**.** Let κ be inaccessible and let $\eta \geq \kappa$ be an ordinal. A partial function f from κ to κ has the *η-strong unfoldability Menas property* if every κ-model M containing f admits an elementary embedding $j : M \to N$, with N transitive, such that

$$\mathrm{crit}(j) = \kappa, \qquad j(\kappa) > \eta, \qquad V_\eta \subseteq N,$$
$$\kappa \in \mathrm{dom}(j(f)), \qquad j(f)(\kappa) \geq (\beth_\eta)^N.$$

The beth number in the last inequality is computed in N. This is the local Menas property at height η of [9, Section 3, p. 1221]. The *strong unfoldability Menas property* requires this at every ordinal height $\eta \geq \kappa$. In this appendix we use only the local property at height $\kappa + 1$.

We use the local embedding data of [9, Fact 5 and Theorem 10(2)] in the following form.

Proposition A.3. *If κ is Π^2_1-indescribable, there is a partial local Menas function f from κ to κ at height $\kappa + 1$ such that, for each κ-model M containing f, one can choose an elementary embedding $j : M \to N$ satisfying*

$$\mathrm{crit}(j) = \kappa, \qquad j(\kappa) > \kappa + 1, \qquad V_{\kappa+1} \subseteq N,$$
$${}^{\kappa}N \subseteq N, \qquad |N| = 2^\kappa, \qquad j(f)(\kappa) > \mu, \qquad \mu = (\beth_{\kappa+1})^N = (2^\kappa)^N.$$

The target regards $j(\kappa)$ as inaccessible and has $V^N_{j(\kappa)}$ as a set.

Proof. Use [9, Corollary 35 and Fact 5] at successor height $\kappa + 1$. Let f_0 be the partial function supplied by [9, Theorem 10(2)], and put $f(\xi) = f_0(\xi) + 1$ for $\xi \in \mathrm{dom}(f_0)$. Its values remain below κ, and every κ-model containing f also contains f_0: take the unique predecessor of each value. As explained in [9, Section 3], the Menas property of f_0 can be retained when selecting the targets of Fact 5. Definition A.2 therefore gives

$$j(f)(\kappa) = j(f_0)(\kappa) + 1 > (\beth_{\kappa+1})^N = \mu.$$

Inaccessibility and external $< \kappa$-closure give $V_\kappa \subseteq M$: by induction on rank, each $x \in V_\kappa$ is the range of a sequence of fewer than κ elements already in M. The same argument puts each V_α, for $\alpha < \kappa$, into M. Replacement then forms V_κ in M, and M regards κ as inaccessible. Elementarity gives inaccessibility of $j(\kappa)$ in N and $j(V_\kappa) = V^N_{j(\kappa)} \in N$. □

A.2. **A presentation determined by H_{κ^+}.** For the preservation argument, we need conditions and side-function names of hereditary size at most κ. Fix a strongly inaccessible κ and write $K = H_{\kappa^+}$, calculated in the ground universe. No assumption on 2^κ is needed here. We use the support conventions of Definition 4.1 and represent each coordinate by a decided stem and a code for a nice function name.

A *nice function-name code* over a forcing order E is a sequence of the fixed length κ,

$$B = \langle (A_\zeta, v_\zeta) : \zeta < \kappa \rangle,$$

where A_ζ is a maximal antichain in E and $v_\zeta : A_\zeta \to \kappa$. It determines the name

$$\dot{h}_B = \{ (\langle \zeta, \check{v_\zeta}(a) \rangle, a) : \zeta < \kappa,\ a \in A_\zeta \}$$

for the graph of a function from κ to κ. The code itself, rather than the expansion of its check names, occurs in the conditions below.

Definition A.4 (Canonical presentation)**.** Define $\mathbb{E}_\theta$, for $\theta \le \kappa^+$, recursively, starting with $\mathbb{E}_0 = \{\varnothing\}$. A condition in $\mathbb{E}_\theta$ is a partial function $p \in K$ whose domain is a subset of θ of size $< \kappa$, such that $p \restriction \xi \in \mathbb{E}_\xi$ for every $\xi < \theta$. For $\xi \in \mathrm{dom}(p)$, its entry has the form

$$p(\xi) = (s_\xi, B_\xi), \qquad s_\xi \in {}^{<\kappa}\kappa,$$

where B_ξ is a nice function-name code over $\mathbb{E}_\xi$ and

$$p \restriction \xi \Vdash_{\mathbb{E}_\xi} s_\xi \subseteq \dot{h}_{B_\xi}.$$

At an omitted coordinate we use the empty stem and the constant zero function, without adding that coordinate to the domain. The empty function is the greatest condition.

For $p, q \in \mathbb{E}_\theta$, define $q \le p$ if $\mathrm{supp}(p) \subseteq \mathrm{supp}(q)$ and, for every $\xi \in \mathrm{supp}(p)$,

$$s^p_\xi \subseteq s^q_\xi \quad \text{and} \quad q \restriction \xi \Vdash_{\mathbb{E}_\xi} \dot{h}_{B^p_\xi} \le \dot{h}_{B^q_\xi} \text{ pointwise.}$$

Every valid entry extends the greatest condition assigned to an omitted coordinate, so no additional comparison is required at a new coordinate.

Lemma A.5. *Each $\mathbb{E}_\theta$ is forcing equivalent to $\mathbb{P}_\theta$, is $< \kappa$-closed and κ^+-Knaster, and has all its conditions in K. Every nice function-name code over $\mathbb{E}_\theta$, and every family of at most κ of its conditions, belongs to K.*

Proof. We induct on $\theta \le \kappa^+$, constructing maps $i_\theta : \mathbb{E}_\theta \to \mathbb{P}_\theta$ that preserve and reflect order, preserve incompatibility, and have dense range. They will also satisfy

$$\mathrm{supp}(i_\theta(p)) = \mathrm{supp}(p), \qquad i_\theta(p) \restriction \xi = i_\xi(p \restriction \xi) \quad (\xi < \theta). \tag{7}$$

For $\theta = 0$, use the identity map. The only condition is empty, and every nice code has hereditary size at most κ. Suppose now that $\theta > 0$ and that the lemma and the maps have been established at all earlier stages.

Hereditary size. Write $T(x) = \mathrm{tc}(\{x\})$. If $\xi < \theta$ and $B = \langle (A_\zeta, v_\zeta) : \zeta < \kappa \rangle$ is a nice code over $\mathbb{E}_\xi$, the induction hypothesis gives

$$|A_\zeta| \le \kappa, \qquad |T(a)| \le \kappa \quad (a \in A_\zeta).$$

The ordinal indices and values contribute at most κ elements, and coding sequences and ordered pairs adds only finitely many components per entry. Hence

$$|T(B)| \le \kappa + \sum_{\zeta<\kappa} \sum_{a \in A_\zeta} (\aleph_0 + |T(a)|) \le \kappa. \tag{8}$$

For a partial function p with domain $u \in [\theta]^{<\kappa}$ and entries of the required form, each $\xi \in u$ is below κ^+, so $|T(\xi)| \le \kappa$. Its stems have hereditary size less than κ. Thus

$$\begin{aligned} |T(p)| &\le \aleph_0 + \sum_{\xi \in u} \big(\aleph_0 + |T(\xi)| + |T(s_\xi)| + |T(B_\xi)|\big) \\ &\le \aleph_0 + |u| \cdot \kappa \le \kappa. \end{aligned} \tag{9}$$

In particular, the requirement $p \in K$ excludes no partial function with the prescribed entries and support size. These estimates apply to the conditions constructed below.

The map and its order. At every earlier stage ξ, translate $\mathbb{E}_\xi$-names recursively by

$$\tau^{i_\xi} = \{(\rho^{i_\xi}, i_\xi(a)) : (\rho, a) \in \tau\}.$$

Forcing equivalence at that stage gives

$$r \Vdash_{\mathbb{E}_\xi} \varphi(\tau) \quad \Longleftrightarrow \quad i_\xi(r) \Vdash_{\mathbb{P}_\xi} \varphi(\tau^{i_\xi}). \tag{10}$$

For a nice code, the translated graph is, up to forced equality of check names,

$$\dot{h}^{i_\xi}_B = \{(\langle \zeta, \check{v_\zeta}(a) \rangle, i_\xi(a)) : \zeta < \kappa,\ a \in A_\zeta\}.$$

For $p \in \mathbb{E}_\theta$, give $i_\theta(p)$ the same domain as p. At a coordinate ξ in this domain, validity and (10) show that

$$i_\xi(p \restriction \xi) \Vdash_{\mathbb{P}_\xi} \langle \check{s}^p_\xi, \dot{h}^{i_\xi}_{B^p_\xi} \rangle \in \dot{\mathbb{Q}}_\xi.$$

Fullness of the iterand name therefore supplies a representative $i_\theta(p)(\xi)$ in its domain with

$$i_\xi(p \restriction \xi) \Vdash_{\mathbb{P}_\xi} i_\theta(p)(\xi) = \langle \check{s}^p_\xi, \dot{h}^{i_\xi}_{B^p_\xi} \rangle. \tag{11}$$

Choose this representative once for each triple $(p \restriction \xi, s^p_\xi, B^p_\xi)$. This makes the choices coherent across stages and gives (7). Coordinate validity and the support bound give $i_\theta(p) \in \mathbb{P}_\theta$.

If $q \leq p$ in $\mathbb{E}_\theta$, induction gives $i_\xi(q \restriction \xi) \leq i_\xi(p \restriction \xi)$ for every $\xi < \theta$. This initial condition forces both coordinate identities in (11). The stem inclusion and side-function inequality defining $q \leq p$ then give the coordinate inequalities for $i_\theta(q) \leq i_\theta(p)$. Conversely, if the latter inequality holds, restriction and order reflection at earlier stages give $q \restriction \xi \leq p \restriction \xi$. Substituting the coordinate identities into the forced Hechler inequality yields stem inclusion and, by (10), the required side-function inequality in $\mathbb{E}_\xi$. The domains are unchanged. Thus

$$i_\theta(q) \leq_{\mathbb{P}_\theta} i_\theta(p) \quad \Longleftrightarrow \quad q \leq_{\mathbb{E}_\theta} p. \tag{12}$$

Density and forcing equivalence. Given $p \in \mathbb{P}_\theta$, choose $p' \in \mathcal{E}_\theta$ below p, using Lemma 4.4, and put $u = \operatorname{supp}(p')$. For $\xi \in u$, let s_ξ be its decided stem. Extend the side-function name to a globally total $\mathbb{P}_\xi$-name $\dot f_\xi$: on a maximal antichain containing $p' \restriction \xi$, retain that name at $p' \restriction \xi$ and use zero at all other members. Then

$$\mathbf{1}_\xi \Vdash \dot f_\xi : \kappa \to \kappa, \qquad p' \restriction \xi \Vdash p'(\xi) = \langle \check s_\xi, \dot f_\xi \rangle.$$

For every $\zeta < \kappa$, conditions $a \in \mathbb{E}_\xi$ whose images decide $\dot f_\xi(\zeta)$ are dense in $\mathbb{E}_\xi$. Indeed, below $i_\xi(r)$, first decide this value in $\mathbb{P}_\xi$, then use the dense range to choose an image below the deciding condition; order reflection puts its preimage below r. Choose a maximal antichain A_ζ of such conditions and record the decided values in $v_\zeta : A_\zeta \to \kappa$. Since i_ξ preserves incompatibility and has dense range, $i_\xi[A_\zeta]$ is a maximal antichain in $\mathbb{P}_\xi$. More explicitly, for any $t \in \mathbb{P}_\xi$, choose $i_\xi(w) \leq t$; maximality in $\mathbb{E}_\xi$ supplies a common extension of w and some $a \in A_\zeta$, whose image witnesses compatibility with t. Therefore the code $B_\xi = \langle (A_\zeta, v_\zeta) : \zeta < \kappa \rangle$ satisfies

$$\mathbf{1}_\xi \Vdash_{\mathbb{P}_\xi} \dot h^{i_\xi}_{B_\xi} = \dot f_\xi.$$

The earlier chain condition and (8) give $B_\xi \in K$.

Define q on u by $q(\xi) = (s_\xi, B_\xi)$. Inductively, suppose $i_\xi(q \restriction \xi) \leq p' \restriction \xi$. For $\xi \in u$, this condition forces $s_\xi \subseteq \dot h^{i_\xi}_{B_\xi}$, so (10) verifies the entry's validity in $\mathbb{E}_{\xi+1}$. It also forces the new coordinate of the image to equal $p'(\xi)$, by (11). Hence

$$i_{\xi+1}(q \restriction (\xi+1)) \leq p' \restriction (\xi+1).$$

At omitted coordinates this inequality is unchanged. At a limit, all preceding validity tests hold and the domain remains a subset of u, of size $< \kappa$. The hereditary-size estimate consequently gives

$$q \in \mathbb{E}_\theta, \qquad \operatorname{supp}(q) = u, \qquad i_\theta(q) \leq p' \leq p.$$

Thus the range of i_θ is dense.

If two images have a common extension, density gives an image below it, and order reflection gives a common extension of their preimages. The converse follows from order preservation. In particular, i_θ preserves incompatibility. The associated generic filters correspond by

$$H = \{r \in \mathbb{P}_\theta : (\exists p \in G)\ i_\theta(p) \leq r\}, \qquad G = \{p \in \mathbb{E}_\theta : i_\theta(p) \in H\}.$$

For a generic H, there is an image in H below every member of H: the images below a given condition, together with the conditions incompatible with it, form a dense set. This fact and order reflection give directedness of the second filter and show that the assignments are inverse. The image of a dense subset of $\mathbb{E}_\theta$ is dense in $\mathbb{P}_\theta$; for a dense $D \subseteq \mathbb{P}_\theta$, the set of $p \in \mathbb{E}_\theta$ whose images extend a member of D is dense. Thus both filters are generic. Induction on name rank gives $\tau^G = (\tau^{i_\theta})^H$. This proves forcing equivalence and establishes (10) at stage θ.

Apply the κ^+-Knaster property of $\mathbb{P}_\theta$, from Lemma 4.4, to the images of a κ^+-sequence in $\mathbb{E}_\theta$. Compatibility of images is equivalent to compatibility of preimages, so $\mathbb{E}_\theta$ is κ^+-Knaster. Its antichains therefore have size at most κ, and (8) applies also to nice codes over $\mathbb{E}_\theta$. If $X \subseteq \mathbb{E}_\theta$ and $|X| \leq \kappa$, then

$$|T(X)| \leq \aleph_0 + \sum_{p \in X} (1 + |T(p)|) \leq \kappa.$$

This proves the remaining hereditary-size assertions.

Closure. Let $\langle p_i : i < \delta \rangle$ be decreasing in $\mathbb{E}_\theta$, where $0 < \delta < \kappa$, and put

$$u = \bigcup_{i<\delta} \operatorname{supp}(p_i).$$

Regularity of κ gives $|u| < \kappa$. For $\xi \in u$, denote the coordinate stems and side-function names by s^i_ξ and $\dot h^i_\xi$, using the empty stem and zero function when ξ is omitted. The stems are increasing, so

$$t_\xi = \bigcup_{i<\delta} s^i_\xi \in {}^{<\kappa}\kappa.$$

The earlier $< \kappa$-closure preserves regularity of κ. As each side-function name is globally total, we may choose a name $\dot g_\xi$ with

$$\mathbf{1}_{\mathbb{E}_\xi} \Vdash \dot g_\xi(\zeta) = \sup_{i<\delta} \dot h^i_\xi(\zeta) < \kappa \quad (\zeta < \kappa).$$

Choose maximal antichains deciding these values and let C_ξ be the resulting nice code. The earlier chain condition gives

$$C_\xi \in K, \qquad \mathbf{1}_{\mathbb{E}_\xi} \Vdash \dot h_{C_\xi} = \dot g_\xi.$$

Define q on u by $q(\xi) = (t_\xi, C_\xi)$. Suppose inductively that $q \restriction \xi \le p_i \restriction \xi$ for all $i < \delta$. Below this initial condition, the side functions increase with i, and each extends its stem. If $\zeta < \mathrm{dom}(t_\xi)$, choose $i_0 < \delta$ with $\zeta < \mathrm{dom}(s^{i_0}_\xi)$. For every $j \ge i_0$, stem extension and normalization give $\dot h^j_\xi(\zeta) = t_\xi(\zeta)$; for $j < i_0$, the forced side-function inequality gives $\dot h^j_\xi(\zeta) \le t_\xi(\zeta)$. Consequently

$$q \restriction \xi \Vdash \dot h_{C_\xi}(\zeta) = \sup_{j<\delta} \dot h^j_\xi(\zeta) = t_\xi(\zeta).$$

This verifies normalization of the new entry. Since t_ξ extends every s^i_ξ and $\dot h_{C_\xi}$ pointwise dominates every $\dot h^i_\xi$, the same initial condition verifies

$$q \restriction (\xi+1) \le p_i \restriction (\xi+1) \quad (i < \delta).$$

At limits, the domain has size $< \kappa$ and all earlier validity tests remain satisfied. By (9), $q \in K$, so $q \in \mathbb{E}_\theta$ and $q \le p_i$ for all $i < \delta$. The empty condition bounds an empty sequence. This proves closure and completes the induction. □

Lemma A.6. *Suppose $W \subseteq U$ is a transitive model of $\mathsf{ZFC}^-_{\mathrm{Coll}}$, with the correct κ^+, and*

$$H^W_{\kappa^+} = H^U_{\kappa^+} = K \in W.$$

Then

$$(\mathbb{E}_\theta)^W = (\mathbb{E}_\theta)^U \qquad (\theta \le \kappa^+),$$

including the order relations.

Proof. We describe the recursion using tests on sets that are common to the two models. All candidate conditions lie in K, as do their restrictions, stems, and codes. For such a candidate, the assertion that its support has size $< \kappa$ is absolute. An external bijection from some $\delta < \kappa$ onto that support has hereditary size at most κ, hence belongs to $K \subseteq W$; conversely, an internal bijection remains a bijection in U.

Suppose the earlier conditions and orders are common. Compatibility in $\mathbb{E}_\xi$ is then absolute, since it asks for a common extension in that same set. The statement that a given A is a maximal antichain also quantifies over the common set $\mathbb{E}_\xi$. Thus the nice-code tests agree. The forcing assertions in Definition A.4 can be expressed directly in terms of these codes. For example,

$$r \Vdash \dot h_B(\zeta) = \check\gamma \quad \Longleftrightarrow \quad (\forall a \in A_\zeta)\big(\mathrm{Comp}(r,a) \Rightarrow v_\zeta(a) = \gamma\big).$$

Maximality ensures that the compatible members decide all possible values below r. Similarly, $r \Vdash \dot h_B(\zeta) \le \dot h_C(\zeta)$ holds exactly when there are no $a \in A^B_\zeta$, $b \in A^C_\zeta$ with $v^B_\zeta(a) > v^C_\zeta(b)$ and a common extension of r, a, b. These tests quantify over the common earlier order. Testing equality at each position in a stem gives normalization; testing the inequality at every $\zeta < \kappa$ gives the pointwise comparison. Stem inclusion is absolute as well.

At stage θ, Separation on K gives the conditions satisfying these tests, and Separation on $K \times K$ gives their order. These fixed sets belong to W, and the tests define a unique next stage from each earlier sequence. Transfinite recursion using Replacement therefore collects the stages through κ^+ inside W. Here κ^+ is a parameter in W, while all coordinates appearing in conditions are below κ^+. Induction gives equality at every stage, including limits: the support-size test places no boundedness requirement on the domain. □

A.3. **Agreement with the prepared universe.**

Lemma A.7. *Suppose N is transitive and satisfies $\mathsf{ZFC}^-_{\mathrm{Coll}}$, $V_{\kappa+1} \subseteq N$, and N has an internally inaccessible $\nu > \kappa$ with $V_\nu^N \in N$. Then*

$$(\kappa^+)^N = \kappa^+, \qquad H^N_{\kappa^+} = H_{\kappa^+}, \qquad H_{\kappa^+} \in N.$$

Proof. The rank segment V_ν^N supplies the internal sets $\mathcal{P}(\kappa)^N$, $(\kappa^+)^N$, and $H^N_{(\kappa^+)^N}$. Since $V_{\kappa+1} \subseteq N$, every actual subset of κ belongs to N. Fix in N a bijection from $\kappa \times \kappa$ to κ. Every well-order of a subset of κ is therefore coded by a subset of κ in N, and its Mostowski collapse in N is its actual collapse. Thus every $\alpha < \kappa^+$ belongs to N and has an internal enumeration of length at most κ. It follows that $(\kappa^+)^N \geq \kappa^+$. Strict inequality would give an internal enumeration of the actual κ^+ by κ, also an enumeration in the ambient universe. Hence $(\kappa^+)^N = \kappa^+$.

For $x \in H_{\kappa^+}$, code $\mathrm{tc}(\{x\})$ by a well-founded extensional relation on a subset of κ, with a distinguished element corresponding to x. This code belongs to N. Its internal collapse gives x and an enumeration witnessing its hereditary size at most κ. Conversely, any such internal enumeration is an actual enumeration. Therefore $H^N_{\kappa^+} = H_{\kappa^+}$; the common set belongs to N because the internal set exists. □

Lemma A.8. *Let $j : M \to N$ be as in Proposition A.3, and let $G \subseteq \mathbb{R}$ be generic over V. Put $U = V[G]$. Then*

$$H^{N[G]}_{\kappa^+} = H^U_{\kappa^+} \in N[G].$$

If $V \models 2^\kappa = \kappa^+$, then $U \models 2^\kappa = \kappa^+$.

Proof. The preparation below κ is the same in N and V, so G is also N-generic. Every relation on κ in U has a nice $\mathbb{R}$-name with at most κ pairs. The conditions in this name belong to V_κ, so the name has hereditary size at most κ. By Lemma A.7, it belongs to N. Consequently $N[G]$ contains every relation on κ in U.

The κ-cc of $\mathbb{R}$ preserves κ^+ both in V and internally in N. Internally, $\mathbb{R}$ has size $\kappa < j(\kappa)$, so $j(\kappa)$ remains inaccessible in $N[G]$, and $V^{N[G]}_{j(\kappa)} \in N[G]$. The coding and collapse argument of Lemma A.7 now proves the assertion about H_{κ^+}.

Finally, if $2^\kappa = \kappa^+$ in V, the number of nice names for subsets of κ is at most $(2^\kappa)^\kappa = \kappa^+$. Together with preservation of κ^+ and the old subsets of κ, this gives $U \models 2^\kappa = \kappa^+$. □

The proof uses the external size $|N|^V = \kappa^+$ to enumerate dense sets. The lottery bound, however, is the internal cardinal $\mu = (2^\kappa)^N$. We do not assume that $N \models 2^\kappa = \kappa^+$.

A.4. **Lifting the embedding.**

Proof of Theorem 5.1. Fix $\theta \leq \kappa^+$. In the prepared universe use $Q = \mathbb{E}_\theta$, forcing equivalent to $\mathbb{P}_\theta$ by Lemma A.5, and let $\dot{Q}$ be its canonical $\mathbb{R}$-name. In V, fix an $\mathbb{R} * \dot{Q}$-name $\dot{A}$ forced to be a subset of κ, and an arbitrary condition r. Choose

$$X \prec H_\Lambda, \quad |X| = \kappa, \quad {}^{<\kappa}X \subseteq X, \quad \kappa + 1 \subseteq X,$$

with $\kappa, f, \mathbb{R}, \dot{Q}, \dot{A}, r \in X$, where Λ is sufficiently large for the forcing constructions under consideration. Since $\mathbb{R}$ is κ-cc and forces that Q is κ^+-cc, $\mathbb{R} * \dot{Q}$ is κ^+-cc. Every maximal antichain in X consequently has size at most κ and is contained in X: an enumeration by κ can be chosen in X, and $\kappa \subseteq X$. Thus the restriction to X of any actual generic filter $G * g$ containing r is X-generic. Here G is V-generic for $\mathbb{R}$, and g is U-generic for the full order Q, where $U = V[G]$.

Let $\pi : X \to M$ be the transitive collapse. The choice of X makes M a κ-model. Since $V_\kappa \subseteq X$, the map π fixes $\kappa, f, \mathbb{R}$ and every condition of $\mathbb{R}$. Put

$$\dot{Q}_0 = \pi(\dot{Q}), \qquad \dot{A}_0 = \pi(\dot{A}).$$

By [9, Lemma 19], the image of $(G * g) \cap X$ is an M-generic filter $G * g_0$ on $\mathbb{R} * \dot{Q}_0$, and

$$A = \dot{A}^{G*g} = \dot{A}_0^{G*g_0} \in M[G][g_0].$$

Writing $Q_0 = \dot{Q}_0^G$, we have $M[G] \models Q_0$ is $< \kappa$-closed. This collapsed order may differ from Q; the filter g_0 will be used in the domain, while the actual filter g will be used in the target.

Choose $j : M \to N$ as in Proposition A.3. Lemmas A.8 and A.6 give

$$Q = (\mathbb{E}_\theta)^U = (\mathbb{E}_\theta)^{N[G]} \in N[G].$$

Indeed, $N[G]$ has the correct κ^+ and contains the common set H_{κ^+} as an element. In particular, $\kappa^+ < j(\kappa)$.

We next verify the internal size bound needed for the lottery. The size-κ, κ-cc preparation preserves the internal cardinal $\mu = (2^\kappa)^N$ and its value as 2^κ, by the nice-name count in Lemma A.8. Internally in $N[G]$, H_{κ^+} has cardinality $2^\kappa = \mu$: its elements are coded by relations on κ, and it contains all subsets of κ. Every condition of Q belongs to this transitive set, and the order is a relation on its underlying set. Hence

$$N[G] \models |\operatorname{tc}(\{Q\})| \leq \mu \quad \text{and} \quad Q \text{ is } < \kappa\text{-closed}.$$

Since $j(f)(\kappa) > \mu$, the actual order Q belongs to the stage-κ lottery of $j(\mathbb{R})$. This stage is eligible because $\kappa \in \operatorname{dom}(j(f))$ and $j(f)''\kappa = f''\kappa \subseteq \kappa$. Let $\dot{Q}_N$ be the canonical $\mathbb{R}$-name for $\mathbb{E}_\theta$ formed in N. The internal size and closure calculations hold in every $\mathbb{R}$-generic extension of N, so the greatest condition forces that $\dot{Q}_N$ is eligible for this lottery. Choose z with trivial restriction below κ and with stage-κ coordinate choosing $\dot{Q}_N$. Since $\dot{Q}_N^G = Q$, below z the image preparation factors as

$$j(\mathbb{R}) \restriction z \cong \mathbb{R} * \dot{Q}_N * \mathbb{R}_{\text{tail}}.$$

Every later eligible stage lies above $j(f)(\kappa)$. The closure of the iterands and Easton support therefore make the tail internally closed under descending sequences of length at most μ, as in [9, p. 1229].

First lift. Work in $T = V[G][g]$. Apply [9, Fact 14(3)] twice, with $\delta = \kappa^+$. Initially ${}^\kappa N \subseteq N$; the two forcings are κ^+-cc, and their conditions belong to the successive targets: $\mathbb{R} \subseteq N$ and $Q \subseteq H^U_{\kappa^+} \subseteq N[G]$. Thus

$${}^\kappa(N[G][g]) \subseteq N[G][g] \quad \text{in } T.$$

Evaluation of the names in N gives $|N[G][g]|^T \leq \kappa^+$. In fact equality holds, since $N \subseteq N[G][g]$ has size κ^+ and the two forcings preserve κ^+.

List the dense subsets of $\mathbb{R}_{\text{tail}}$ that belong to $N[G][g]$ in a sequence of length at most κ^+, and build a descending chain meeting them. At a limit $0 < \alpha < \kappa^+$, choose in the target a surjection from κ onto α. Such a surjection exists because the target has the correct κ^+. The preceding conditions, listed along that surjection, form a κ-sequence in the target by external closure. Using the surjection, the target reconstructs the original α-sequence. Since $\alpha < \kappa^+ \leq \mu$, internal tail closure provides a lower bound. No lower bound for the entire κ^+-sequence is needed. The chain generates an $N[G][g]$-generic filter G_{tail} in T.

Under the displayed factorization, let $j(G) = G * g * G_{\text{tail}}$. This is an N-generic filter on $j(\mathbb{R})$ containing $j''G = G$, so the lifting criterion gives

$$j : M[G] \longrightarrow N[j(G)].$$

The target $N[j(G)]$ still has external size at most κ^+. Since $G_{\text{tail}} \in T$, [9, Fact 14(1)], applied in T, also preserves its external κ-sequence closure.

Second lift. The κ-cc preparation preserves the external $< \kappa$-closure of $M[G]$, by [9, Fact 14(3)] with $\delta = \kappa$. The actual order Q is $< \kappa$-distributive, so $M[G]$ remains $< \kappa$-closed in T. Finally, [9, Fact 14(1)], applied in T to the $M[G]$-generic filter g_0, shows that $M[G][g_0]$ is externally $< \kappa$-closed. Since $|M| = \kappa$, evaluating the names in M gives $|M[G][g_0]| \leq \kappa$. The reverse inequality follows from $\kappa \subseteq M[G][g_0]$, so this model has size κ and is a κ-model in T.

We need a descending cofinal chain in g_0. First, every descending sequence of fewer than κ members of g_0 has a lower bound in g_0. To prove this, external closure places the sequence in $M[G][g_0]$. Internal $< \kappa$-distributivity of Q_0 then places it in $M[G]$, since its entries belong to the old set Q_0. For each entry q, the set

$$\{p \in Q_0 : p \leq q \text{ or } p \perp q\}$$

is dense open. Their intersection is dense in $M[G]$ by distributivity. A member of its intersection with g_0 is compatible with every entry, and hence extends every entry.

The filter g_0 is directed and has external size at most κ. Enumerating its members and using the preceding closure at limit stages, choose a descending cofinal chain $\langle q_\xi : \xi < \kappa \rangle$ in g_0. External κ-closure of $N[j(G)]$ puts $\langle j(q_\xi) : \xi < \kappa \rangle$ in that target. By elementarity, $N[j(G)] \models j(Q_0)$ is $< j(\kappa)$-closed. Since $\kappa < j(\kappa)$, there is a condition $q^* \in j(Q_0)$ below the entire image chain, and therefore below every member of $j''g_0$.

Starting below q^*, diagonalize through the dense subsets of $j(Q_0)$ in $N[j(G)]$. There are at most κ^+ such sets. The target still has the correct κ^+: its old enumerations of ordinals below κ^+ remain, and any enumeration of the actual κ^+ by κ in the transitive target would also be one in T. At each limit below

κ^+, the same sequence-coding argument as in the first lift puts the preceding chain in the target; its internal $< j(\kappa)$-closure supplies a lower bound, since $\kappa^+ < j(\kappa)$. This produces in T an $N[j(G)]$-generic filter h with $q^* \in h$. Consequently $j''g_0 \subseteq h$, and the embedding lifts to

$$j^* : M[G][g_0] \longrightarrow N[j(G)][h].$$

Strength of the lifted embedding. Since κ is inaccessible in U and Q is $< \kappa$-closed, $V_\kappa^T = V_\kappa^U$. Fix in U an enumeration of V_κ^U of length κ. Each subset of V_κ^U in T has a nice Q-name with at most κ pairs: choose one antichain deciding membership for each entry of the enumeration, and use the κ^+-cc. Every condition in the name has hereditary size at most κ, by Lemma A.5. The enumeration, the check names for its entries, and the nice name itself therefore have hereditary size at most κ. They belong to the common set $H^U_{\kappa^+} = H^{N[G]}_{\kappa^+}$, and their evaluation by the actual filter g gives

$$V^T_{\kappa+1} \subseteq N[G][g] \subseteq N[j(G)][h].$$

Moreover $j^*(\kappa) = j(\kappa) > \kappa + 1$, and A belongs to the transitive domain of size κ. Since $\dot{A}$ was arbitrary, [9, Fact 4(5) and Corollary 35] give Π^2_1-indescribability in T. The construction works for every generic filter containing the arbitrary condition r, and for every $\theta \leq \kappa^+$, proving the theorem. □

Remark A.9. The initial equality $2^\kappa = \kappa^+$ can be arranged by set forcing. Present $\mathrm{Coll}(\kappa^+, 2^\kappa)$ as the order of partial functions from κ^+ to the old 2^κ of size $< \kappa^+$, ordered by extension. This order is $< \kappa^+$-closed; the union of the generic filter is a surjection onto the old 2^κ. It preserves κ^+ and adds no ordinal sequences of length at most κ, hence no subsets of κ. Thus $2^\kappa = \kappa^+$ in the extension.

To verify preservation of indescribability, note that $V_{\kappa+1}$ and all relation codes on κ are unchanged. Any κ-model in the extension has a code by a well-founded extensional relation on κ. This code is old, and its absolute collapse is a ground-model κ-model: the old code already witnesses its size, its theory is absolute, and its closure in the extension includes all old short sequences. An old $(\kappa + 1)$-strong unfoldability embedding for that model still has the required rank inclusion and is still elementary. Hence Π^2_1-indescribability is preserved; compare [9, Lemma 6]. One may then apply Theorem 5.1. Our consistency proof starts in L, where this preliminary collapse is unnecessary.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

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